Answering questions with data

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This comprehensive resource offers a free, accessible textbook for environmental science students embarking on introductory statistics. The package includes a practical lab manual and a dedicated course website, all provided under a CC BY-SA 4.0 license.

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Preface

Second Draft (version 0.1 = August 18th, 2025)

Welcome to the second edition of this Open Educational Resource (OER) textbook, specifically adapted for Environmental Science students enrolled in the SPEA E-538 statistics course at Indiana University (IU).

This textbook is an adaptation of a thorough introductory statistics textbook originally developed for undergraduate Psychology students by Matthew Crump and colleagues (refer to Acknowledgements for more details). As part of IU's Course Materials Fellowship Program (CMFP), I've had the opportunity to mold this material, refining it to serve as a specialized resource for students studying Environmental Science.

Online Textbook:https://malloryb.github.io/statistics_E538/

Citation for original textbook: Crump, M. J. C., Navarro, D. J., & Suzuki, J. (2019, June 5). Answering Questions with Data (Textbook): Introductory Statistics for Psychology Students. https://doi.org/10.17605/OSF.IO/JZE52

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My heartfelt thanks also go to the Course Materials Fellowship Program (CMFP) at Indiana University (IU), which has been instrumental in supporting this adaptation effort. The CMFP is an initiative designed to incentivize the discovery, implementation, and creation of cost-effective course materials. Its aim is to foster the use of Open Educational Resources (OERs)—freely accessible and customizable learning materials that make education more equitable and accessible.

I am particularly grateful to Sarah Hare, the Bloomington lead for the CMFP program, and Adam Mazel, a digital publishing librarian at IU. Their guidance, expertise, and steadfast

support have been crucial to this project's success. Their contributions to the advancement of affordable and accessible education are truly commendable.

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Copying the textbook

This textbook was written in R-Studio, using R Markdown, and compiled into a web-book format using the bookdown package. In general, I thank the larger R community for all of the amazing tools they made, and for making those tools open, so that I could use them to make this thing.

All of the source code for compiling the book is available from the GitHub repository for the original textbook:

https://github.com/CrumpLab/statistics

and my github repository:

https://github.com/malloryb/statistics_E538

In principle, anyone can fork or download the E-538 textbook, or the original textbook, which is what I did. You can load the Rproj file in RStudio, compile the entire book, and then edit the individual .Rmd files for content and style to fit your needs.

If you'd like to contribute to this version, you can submit pull requests on GitHub or use the issues tab to share suggestions.

The vision behind this textbook

The aim of this textbook is twofold: to make core statistical concepts accessible to Environmental Science students, and to promote the use of open-source tools like R as flexible, transparent resources for learning and research.

1 Why Statistics?

Adapted to environmental science by Mallory Barnes. Portions adapted from Chapters 1 and 2 in Navarro, D. J. "Learning Statistics with R." https://compcogscisydney.org/learning-statistics-with-r/

> To call in statisticians after the experiment is done may be no more than asking them to perform a post-mortem examination: They may be able to say what the experiment died of.*
> – Sir Ronald Fisher

1.1 The Role of Statistics in Environmental Science

Many students come into statistics classes with a mix of nerves and low expectations. It is not always the most eagerly anticipated part of an environmental science degree. But statistics is one of the most important tools we have for understanding complex systems. Environmental data are messy: ecosystems shift, climate varies, and our judgments are often biased. Without statistical methods we risk telling convenient stories. With them, we can detect real patterns, test competing explanations, and make stronger arguments for environmental decisions.

1.1.1 Why Numbers Matter

Common sense is useful, but it is prone to bias. When we already believe something to be true, we are more likely to interpret new evidence as supporting it, even if it does not. This tendency can lead us astray in environmental science.

For example, if you are convinced that industrial farming is the main driver of bee declines, you might see every new drop in bee abundance as confirmation of that belief. Maybe you are right, but maybe disease or weather are stronger contributors. Statistics gives us tools to test these ideas systematically. It helps us separate patterns from noise, avoid being misled by our own expectations, and draw conclusions that are more likely to hold up under scrutiny. It's not magic, but it makes our conclusions far more reliable.

Another challenge is that data can tell very different stories depending on how they are aggregated. **Simpson's paradox** occurs when a trend seen in combined data reverses after you split the data into meaningful groups.

Here's a recent example. During the COVID-19 pandemic, early data showed that Italy's overall case fatality rate was higher than China's. But, once researchers *disaggregated by age group,* the trend flipped: within every age group, the fatality rate was actually higher in China. The apparent contradiction was explained by differences in the age distribution of cases (von Kügelgen, Gresele, and Schölkopf 2021).

Using case fatality rate (CFR), here are some illustrative numbers (made up for teaching).

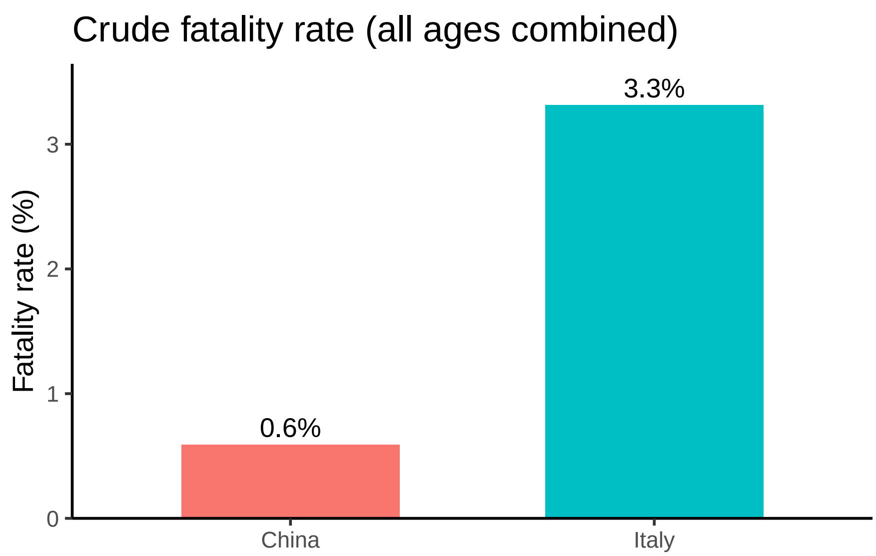

Figure 1.1: Crude case fatality rates. Italy appears to have a higher fatality rate than China when all cases are combined.

At face value, Italy looks worse. But what happens when we stratify by age group? Because age strongly affects COVID fatality, splitting cases into 10-year bands reveals a different picture. When we do, China's fatality rate is higher in every band.

Why the reversal? Italy's cases skew older while China's skew younger, and baseline risk rises steeply with age. The differing age distributions weight the overall averages in opposite ways. Aggregating without age adjustment hid the within-group pattern.

Takeaway

Statistics will not answer every scientific question, but it gives us a disciplined way to sort signal from bias so our inferences age well.

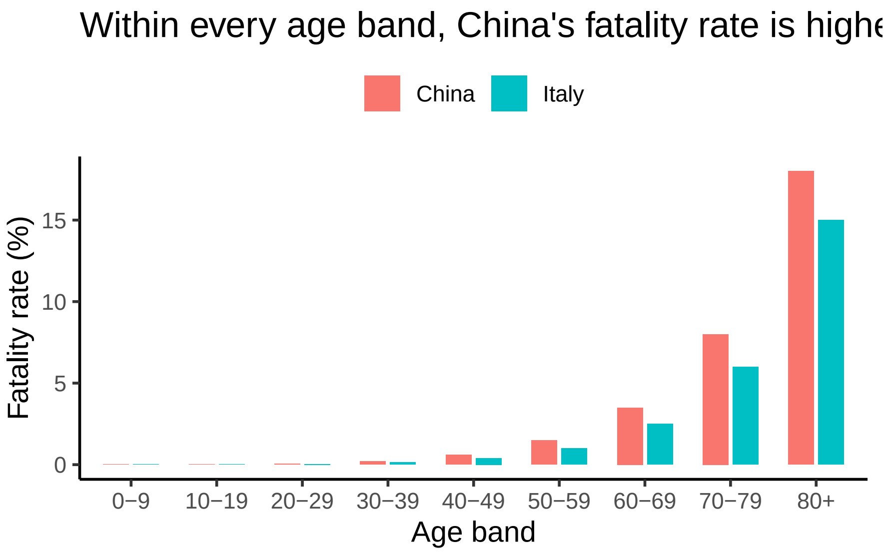

Figure 1.2: Fatality rates by age group (same made-up numbers). Within each age band, China's CFR exceeds Italy's.

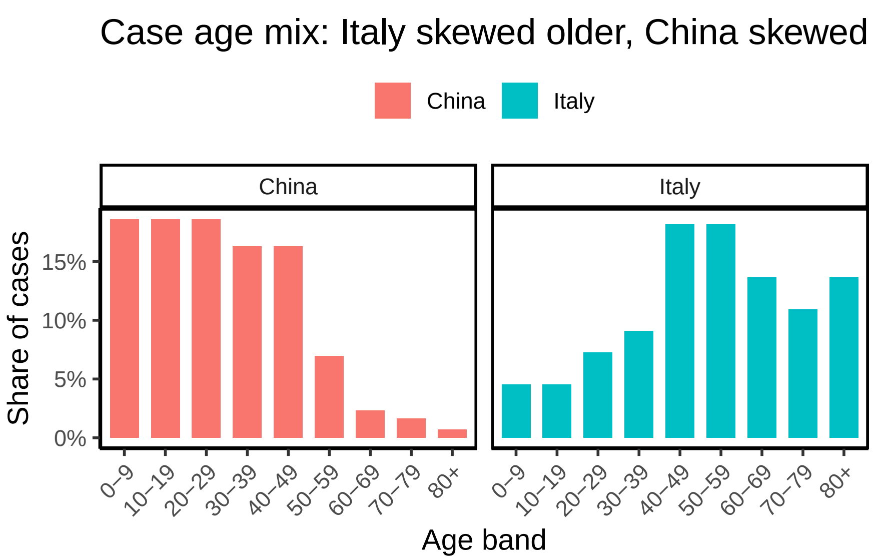

Figure 1.3: Case age distribution by country. Italy's cases are older on average; China's are younger.

1.1.2 What Statistics Add

Environmental science produces a *lot* of data. Every day satellites measure surface temperature, field stations log rainfall, and sensors track air quality. A single study can generate thousands of rows of numbers. Without statistical tools, these numbers would be overwhelming. With them, we can summarize, compare, and test ideas systematically.

Statistics matters here because environmental systems are messy. River flow changes by the hour, forests grow unevenly, and human actions layer on additional variability. Patterns are rarely obvious by eye. Statistics helps us sort signal from noise and ask: is this apparent change real, or just random variation?

You might think, "can't a statistician just handle the math?" But knowing the basics yourself is essential for three reasons:

1. **Design and analysis go together.** A good study starts long before you run a statistical test. If you want to study how fertilizer affects crop yields, your sampling plan and your analysis are inseparable. Poor design, like measuring only in unusually wet fields, can't be rescued by sophisticated analysis.
2. **Understanding the Science:** Scientific papers on climate change, biodiversity, or pollution are built on statistical results. To interpret them, you need to know what the numbers mean.
3. **Practicality:** Hiring a specialist for every question isn't realistic. A working knowledge of statistics makes you more self-sufficient as a scientist.

> *"We are drowning in information, but we are starved for knowledge"*
>
> – Various authors, original probably John Naisbitt

Finally, statistics is not only for researchers. Weather forecasts, air quality alerts, and wildlife population trends are all communicated through numbers. A basic knowledge of statistics helps you tell whether those numbers support the claims being made, or whether someone is stretching the truth. In that sense, statistics is part of being an informed citizen in a data-saturated world.

1.2 Introduction to measurements

Every dataset begins with measurement. Measurement means assigning numbers, labels, or categories to aspects of the world so they can be recorded and analyzed.

Why measurement matters

Environmental science often deals with broad ideas that need to be defined before they can be studied. For example, *soil health* or *forest change* are meaningful, but vague. To analyze them, we have to decide exactly what we mean and how to capture it. That process is called **operationalization**: turning a general concept into something measurable.

Before moving on, let's clarify how operationalization fits with three related terms:

- **Operationalization**: the step where you define exactly how a broad idea will be captured with a measure
- **Measure**: the tool or method used to make observations
- **Variable**: the actual values you record once the measure is applied. Variables are the actual data that end up in our dataset.

Examples:

- Concept of interest: *Soil carbon sequestration*
  - **Operationalization:** concentration of organic carbon in the top 30 cm
  - **Measure:** laboratory analysis of soil samples
  - **Variable:** recorded carbon concentration values for each plot
- Concept of interest: *Deforestation*
  - **Operationalization**: change in forest cover between the current year and five years ago
  - **Measure:** aerial image classification
  - **Variable:** change in forest cover (%) per unit area

Operationalization is rarely straightforward, and there is no single correct way to do it. The best choice depends on the question you are asking and how the data will be used. In many fields, scientists have developed common practices, but every project still requires case-by-case judgment. Even so, some principles of good operationalization apply across studies: be precise about what you mean, how you will measure it, and what values are possible.

1.3 Scales of measurement

As the previous section indicates, the outcome of a measurement is called a **variable**. Not all variables are the same type, and knowing the type matters because it determines which statistical tools make sense. The four classic scales are **nominal, ordinal, interval,** and **ratio**.

1.3.1 Nominal scale

A **nominal scale** variable (also referred to as a **categorical** variable) is one in which the values are just names. They do not have an inherent order, and it makes no sense to average them.

Example: eye color. Eyes can be blue, green, or brown, but none is "greater" than another, and there's no such thing as an "average eye color."

1.3.2 Ordinal scale

Ordinal scale variables have a meaningful order, but the spacing between values is not defined mathematically. You can rank the values, but you can't assume equal steps between them, nor calculate a meaningful average.

Example: finishing position in a race. First place comes before second, and second before third, but you don't know how much faster one runner was than another. Saying the "average place" of the group is 2.3 doesn't mean anything.

1.3.3 Interval scale

Interval variables have equal intervals between values, so differences are meaningful. However, zero is arbitrary, so multiplication and division are not valid.

Example: temperature in degrees Celsius. A difference of 3° means the same regardless of whether it is $7 \to 10$ or $15 \to 18$. But 0° does not mean "no temperature." That zero is defined by the freezing point of water. So while you can say today is 3° warmer than yesterday, you cannot say 20° is "twice as hot" as 10°.

(Alternate example: latitude. A difference of 10° latitude is meaningful, but zero latitude—the equator—doesn't mean "no latitude." Twice 45° is 90°, but that doesn't mean one place has "twice as much latitude" as another.)In contrast to nominal and ordinal scale variables, **interval scale** and ratio scale variables are variables for which the numerical value is genuinely meaningful. In the case of interval scale variables, the *differences* between the numbers are interpretable, but the variable doesn't have a "natural" zero value. You can add and subtract, but ratios don't make sense.

1.3.4 Ratio scale

Ratio scale variables have all the properties of interval variables, plus a true zero. This makes multiplication and division valid, as well as addition and subtraction.

Example: age in years. Zero means no age at all, and someone who is 20 years old really is twice as old as someone who is 10. Differences ($20 - 10 = 10$ years) and ratios ($20 \div 10 = 2$) are both meaningful.

	can rank	can subtract/add	can multiply/divide	example
nominal				eye color
ordinal	x			race position
interval	x	x		temperature (°C)
ratio	x	x	x	age

1.3.5 Continuous versus discrete variables

Another useful distinction is whether a variable can take on values in between others.

- A **continuous variable** can, in principle, take on any value within a range. For example, consider height. If you are 72 inches tall and your friend Cameron is 71 inches tall, Alan could be 71.4 inches and David 71.49 inches. Because we can always imagine a new value in between two others, height is continuous.
- A **discrete variable** is, in effect, a variable that isn't continuous. Discrete variables have separate, distinct values with nothing in between. Nominal variables are always discrete: there is no eye color that falls "between" green and blue in the same way that 71.4 falls between 71 and 72. Ordinal variables are also discrete: although second place falls between first and third, nothing can logically fall between first and second.

Interval and ratio variables can be either. Height (a ratio variable) is continuous. But the number of people living in a household (a ratio variable) is discrete: you cannot have 4.2 people. Temperature in degrees Celsius (an interval variable) is also continuous. But the year you started college (an interval variable) is discrete: there is no year between 2022 and 2023.

The table below shows how the scales of measurement relate to this distinction. Cells with an "x" mark what is possible.

	continuous	discrete
nominal		x
ordinal		x

	continuous	discrete
interval	x	x
ratio	x	x

1.3.6 A note on real data

These categories are guides, not hard rules. For example, survey responses on a 1–5 "strongly disagree" to "strongly agree" scale are technically **ordinal**, but researchers often treat them as "quasi-interval" because the spacing is assumed to be roughly equal.

1.4 Assessing the reliability of a measurement

When we measure something, two questions matter: *is the measurement consistent, and is it accurate?*

- **Reliability** means consistency. If you measure the same thing again, do you get the same result?
- **Validity** means accuracy. Does the measurement reflect the real thing you care about?

These aren't the same. A soil moisture probe that is miscalibrated might always read 5% too high. It is highly **reliable**, you get the same number each time, but not **valid**, since it doesn't reflect true soil moisture. On the other hand, a set of volunteer bird counts might fluctuate from one observer to the next (**low reliability**), but when averaged across many counts, the results may still approximate true abundance (**reasonable validity**).

In practice, a measure that is very unreliable usually ends up being invalid too, because we can't tell which result is right. This is why reliability is often considered a prerequisite for validity.

- Reliability can show up in different ways:
  - **Over time** (*test–retest*): if you sample soil moisture today and tomorrow under the same conditions, do you get the same value?
  - **Across people** (*inter-rater*): if two field crews count birds at the same site, do their tallies agree?
  - **Across tools** (*parallel forms*): if you use two different rain gauges in the same spot, do they record the same rainfall?
  - **Within a test** (*internal consistency*): if multiple survey questions are meant to capture the same attitude, do they give similar answers?

Not every measurement needs (or can even possibly have) every form of reliability. The point is that **reliability is necessary but not sufficient for validity**: a method can be consistently wrong, but if it isn't consistent at all, it's almost impossible to know whether it's right. We'll discuss how we assess validity a bit later in this chapter.

1.5 The role of variables: predictors and outcomes

One last piece of terminology before we leave variables behind. In most studies we have many variables, but when we analyze them, we usually split them into two roles: the **thing we're trying to explain** and the **thing doing the explaining**. To keep it straight, we use Y for the variable being explained, and a X_1, X_2, etc. for the variables used to explain it.

Traditionally, X is called the **indepdendent variable (IV)** and Y is the **dependent variable (DV).** The logic is that if there's a relationship, Y depends on X. These terms can be clunky and can be confusing because: (a) IVs are rarely actually "independent of everything else" and (b) if there's no relationship, then the DV doesn't actually "depend" on the IV at all.

Alternative terminology is often clearer. In experiments, IVs are **manipulations**, and DVs are **measurements:**

role of the variable	classical name	alternative
"to be explained" (Y)	dependent variable (DV)	measurement
"to do the explaining" (X)	independent variable (IV)	manipulation

Another useful pair is **predictors** and **outcomes**. The idea here is that what we use X to make predict or guess about Y:

role of the variable	classical name	alternative
"to be explained" (Y)	dependent variable (DV)	outcome
"to do the explaining" (X)	independent variable (IV)	predictor

1.6 Experimental and non-experimental research

A central distinction in research is between experimental and non-experimental studies. What matters here is the degree of control the researcher has.

In **experimental research**, the researcher deliberately manipulates something (the predictor/IV) and measures its effect on outcomes (the outcome/DV). The goal is to isolate causal effects. To avoid the problem of "something else" influencing the outcome, researchers try to hold other factors constant. In practice, it's almost impossible to identify *everything* that might matter, much less keep it constant. The standard solution is **randomization**. Randomization doesn't eliminate confounds, but it makes them less likely to systematically bias results.

For instance, suppose we wanted to know if smoking causes lung cancer. Observing smokers and non-smokers can only get us so far, because those groups differ in many ways besides smoking, like occupation, income, and diet. A true experiment would require randomly assigning people to smoke or not. You can see how that would be deeply unethical. The same problem comes up in medicine: we know surprisingly little about how certain drugs or exposures affect pregnant people, precisely because we cannot ethically assign them to risky conditions.

In **environmental science**, the limitation isn't usually ethics but feasibility: there's no "control planet" we can set aside as a baseline. Nor can researchers manipulate fundamental drivers like precipitation or temperature. As a result, much of the field relies on **non-experimental research** — quasi-experiments, long-term time series, or detailed case studies that allow scientists to tease out effects in complex systems.

Much environmental science fits a **quasi-experimental design**. For example, researchers may wish to study the effects of industrial pollution on a river system but cannot directly control the pollutants emitted. Instead, they observe existing conditions and make careful comparisons, often using statistical tools to account for confounding variables.

Another important quasi-experimental tool is **time series analysis**, since many environmental processes unfold over decades or centuries. Tracking changes in global temperature or sea level requires years of consistent data, and statistical methods help isolate signals from the noise of natural variability.

Environmental science also makes extensive use of **case studies**, which provide in-depth insights into specific events or locations. A case study might explore the aftermath of a natural disaster, the ecology of a threatened habitat, or the consequences of an environmental policy. While a single case is tied to a particular context, well-designed studies can uncover mechanisms that apply more broadly. For example, a drought in one forest might reveal how water potential and transpiration respond to stress, helping researchers anticipate plant responses in other ecosystems.

1.7 Assessing the validity of a study

Earlier we drew a line between **reliability** (consistency) and **validity** (accuracy). Reliability tells us whether our measurement is stable and repeatable. Validity asks the harder question: are we actually measuring what we think we're measuring, and can we trust the conclusions we draw from it?

1.7.1 Internal validity

Internal validity is about whether the relationship you see is really causal. From an earlier example: if smokers have more lung cancer than non-smokers, does that mean smoking causes cancer? Not necessarily. Smokers may differ from non-smokers in income, occupation, diet, or other ways. These "confounds" muddy causal claims. Experiments with random assignment improve internal validity by balancing out such factors, but in environmental science, we often rely on careful design and statistical controls instead.

1.7.2 External validity

External validity is about generalization. Does what you found in one study hold elsewhere? A soil-warming experiment in one forest plot may not capture how all forests respond. Or water-quality measurements from a single river reach may not represent conditions across the whole watershed. Strong internal validity can still leave you with weak external validity if the setting, population, or conditions are narrow.

1.7.3 Construct validity

Construct validity is about whether your measure really matches the concept. If you care about soil fertility but measure only soil moisture, you have high reliability but weak construct validity: you're consistently measuring the wrong thing. Getting this right requires aligning theory, measurement, and data.

1.8 Confounds, artifacts and other threats to validity

Broadly, the two biggest threats to validity are *confounds* and *artifacts*:

- **Confound**: An additional variable, often unmeasured, that masks or distorts the relationshp between the predictor and the outcome. Confounds threaten *internal validity* because you can't tell whether the predictor really causes the outcome. Two variables are said to be **confounded** if their effects on a response variable cannot be distinguished.
  - **Example:** In a study of tree growth near roads, trees closer to the road might appear to grow faster. But if those same trees also receive more runoff water from pavement, then water availability (not proximity to the road) could explain the growth difference.

- **Artifact**: An aspect of the experimental setup or apparatus that biases the results. An artifact gives the appearance of measuring the phenomenon of interest, but is actually measuring something else introduced by the method itself. Artifacts often undermine *external validity,* and sometimes internal validity too.
  - **Example:** In soft-sediment studies, repeated coring to sample buried organisms disturbs the sediment, exposing animals and disrupting microbial and structural assemblages; later samples then reflect the disturbance caused by earlier sampling rather than true ecological patterns (Skilleter 1996)

Here are some more types of threats to validity:

- **History Effects:** Events outside the study can shift the outcome. Long-term ecosystem monitoring is especially vulnerable: imagine tracking stream chemistry for years, then a wildfire in the watershed alters nutrient fluxes. Your "trend" may just reflect that disturbance rather than the process you meant to study.
- **Repeated Testing Effects:** Even measurements themselves can have an effect. For example, extracting multiple tree-ring cores from the same tree could wound it and reduce growth in later years, influencing the very variable you were trying to measure.
- **Selection bias:** Environmental research often focuses on sites that are easiest to study, like forests near R1 universities, long-established research stations, or accessible watersheds. But those places aren't necessarily representative. If they differ systematically from the broader set of ecosystems, conclusions may not generalize.
- **Differential Attrition:** Long-term field studies depend on cooperation, and dropout can skew results. For example, in a cover-cropping trial, if only the most motivated farmers stick with the practice, the observed benefits won't generalize to all farms.
- **Non-response bias:** Surveys face the same problem: those who answer aren't random. For instance, a survey on attitudes toward conservation easements may mostly attract landowners already supportive of conservation, biasing the results.
- **Regression to the mean:** Extreme cases tend to moderate over time. If you select the fastest-growing trees in a stand for special study, you'll probably find their growth slows later — not necessarily because of your treatment, but simply because growth rates fluctuate and extremes rarely persist.
- **Experimenter bias:** Researchers bring expectations to the field. Choices about where to sample a river, or when to measure soil respiration, can unconsciously favor the expected outcome. The classic cautionary tale is Clever Hans, the horse who "did math" by responding to subtle human cues (Pfungst 1911; Hothersall 2004). The lesson: be aware of how your own expectations can shape results.

- **Reactivity and demand effects:** Sometimes behavior changes because people know they're being observed. A field crew told they are testing the "impact" of a method may record data more carefully than usual, producing results that reflect observer effort rather than the phenomenon itself.

1.8.1 Fraud, deception, and self-deception

> *It is difficult to get a man to understand something, when his salary depends on his not understanding it.*
> – Upton Sinclair

Most scientists are honest, but self-deception is common. Researchers can over-interpret noisy results, design studies that all but guarantee an effect, or (consciously or not) hide inconvenient variables. Publication bias makes it worse: null results often go unreported, so the literature overstates effects. And sometimes, as in Simpson's paradox, aggregated data can mislead if you don't dig deeper. The point isn't to be cynical, it's to recognize that science is done by humans, and humans bring bias, incentives, and blind spots.

1.9 Summary

In this chapter, we have explored essential aspects of research methodology pertinent to environmental statistics:

- **The role of statistics in environmental science:** Statistics does not answer every substantive question, but it provides the discipline to separate signal from noise and make credible inferences from messy data.
- **Operationalization and Measurement**: Defining theoretical constructs and deciding how to measure them is foundational.
- **Scales of measurement and types of variables**: Distinguishing discrete from continuous data, and recognizing nominal, ordinal, interval, and ratio scales.
- **Reliability of a measurement**: If I measure the "same" thing twice, should I expect the same result? In what sense (test–retest, inter-rater, internal consistency)?
- **Terminology: predictors and outcomes**: Can I clearly explain the roles variables play in an analysis (predictor vs. outcome, independent vs. dependent)?
- **Experimental and non-experimental research designs**: Identifying what constitutes an experiment, and how researchers rely on quasi-experiments, time series, and case studies when full control is not possible.

- **Validity and Threats**: Does the study truly measure what it claims to? What pitfalls—such as confounds or artifacts—could bias results?

Study design is a cornerstone of environmental research methodology. While many textbooks provide fuller coverage, e.g., Campbell and Stanley (1963), this chapter highlights the unique challenges of applying these principles in environmental science, where ethical and practical constraints often preclude perfect control. By linking statistics to study design, we see how careful methods make it possible to draw credible inferences from complex, real-world systems.

1.10 Videos

1.10.1 Terms of Statistics

2 Describing Data

> Far better an approximate answer to the right question, which is often vague, than an exact answer to the wrong question, which can always be made precise. *
> —John W. Tukey

Statistics often begins with the problem of **too much data**. Modern environmental science generates vast amounts of information: rainfall at hundreds of stations, thousands of tree measurements in a forest survey, millions of satellite pixels. Looking at raw numbers alone is overwhelming and uninformative. We need tools to compress, summarize, and visualize. These are **descriptive statistics**.

Keep in mind:

1. There is more than one useful way to describe data.
2. You choose a description depending on what helps reveal the pattern you care about.
3. All methods were invented for that reason. They are helpful shortcuts.

Imagine running a survey of 500 people's happiness levels. The full dataset is just a long list of values. Staring at hundreds of numbers tells us little. How happy are most people? How unhappy are the unhappiest? Raw data alone can't answer, we need to **summarize.**

2.1 Looking at the data

2.1.1 Scatter of raw values

One first step is simply to **plot every value** instead of listing them. Let's look at them!

Figure **??** shows 500 measurements of happiness. The horizontal axis is an index of participants (1–500), and the vertical axis shows their reported happiness. Each dot represents one person's score.

The way we choose to plot data makes some patterns easier to see and others harder. What do we notice here? There are indeed 500 dots, spread across a wide range: some values climb as high as about 1500, others fall below –1500. Most points cluster somewhere near zero, with fewer at the extremes.

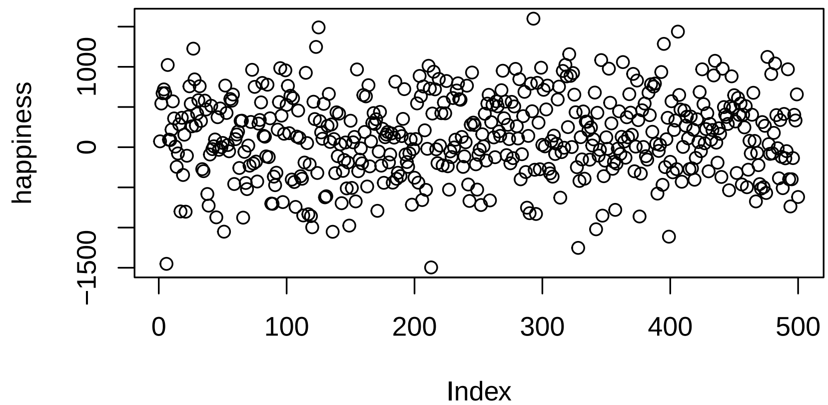

Figure 2.1: Pretend happiness ratings from 500 people

Take-home: plotting all the values at once is already much more useful than staring at a raw table of numbers.

Still, the scatter doesn't make it easy to answer questions like "are most people generally happy or unhappy?" For that, we need a different view, a **histogram**.

2.1.2 Histograms

Making a histogram summarizes data by grouping numbers rather than looking at individual data points. Figure **??** displays 500 happiness scores grouped into bins. Each bar shows how many people's responses fall within a range. The bar between 0 and 500 indicates about 150 people in that range. The y-axis shows the frequency count of each bar.

The histogram makes patterns clearer than the scatterplot. Most of the responses fall between –500 and +500, with only a few extreme highs or lows. We can also see the overall spread of the data: roughly –1500 to +1500.

When making histograms, the width of the bins matters. If the bins are wide, the plot looks smooth but may hide detail. If they are narrow, the plot shows more variation but can appear noisy. Figure **??** compares the same data with different bin widths.

All of the histograms show the same general shape: few values at the extremes and many near zero. Narrow bins make the bars jump up and down, adding noise. Wider bins smooth the pattern, but if they're too wide, important variation gets lost.

Figure 2.2: A histogram of the happiness ratings

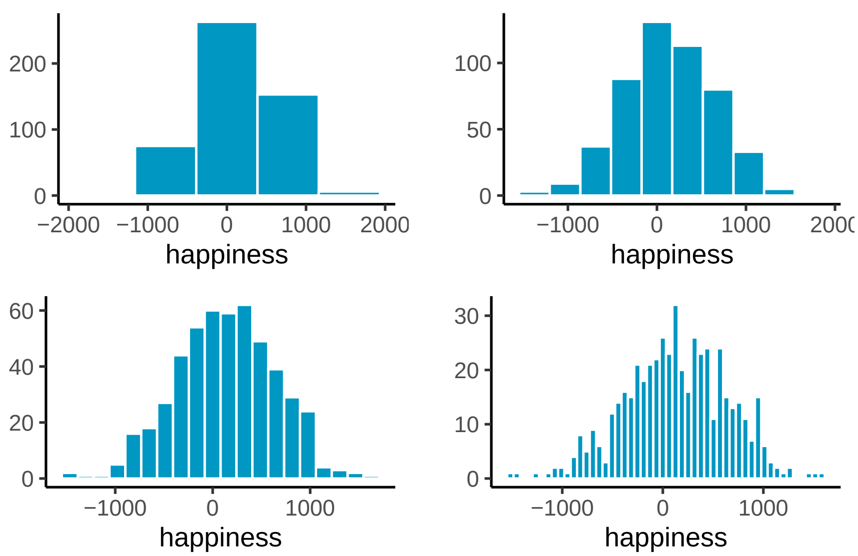

Figure 2.3: Four histograms of the same data using different bin widths

Take-home: histograms summarize individual values into a distribution, revealing both the typical range and how the data are spread. Bin width is a choice you make, and it affects what you see.

2.2 Important Ideas: Distribution, Central Tendency, and Variance

Three terms will keep coming up: **distribution, central tendency, and variance.** Their everyday meanings aren't far from the statistical ones.

- **Distribution**: how values are spread across their range. A histogram is one way of showing a distribution. The shape of the distribution tells us where values are concentrated, where they're rare, and how far the extremes reach. Later we'll see that distributions don't just describe data, we also use them to **generate theoretical data**, which is the basis for sampling distributions and tools like t-tests.
- **Central tendency** is about *sameness*: where values cluster. In the happiness histogram, most responses are near zero, so we say the data have a central tendency around zero. Central tendency doesn't mean "everything is the same", just that many values gather in a common area. Some datasets even have more than one cluster, and therefore more than one central tendency.
- **Variance** is about *differentness*: how spread out the values are. If all responses were nearly identical, variance would be low. In our happiness example, values range widely from strongly negative to strongly positive, so variance is high. In practice, variance is a family of measures we use to quantify differences, and we'll introduce those next.

2.3 Measures of Central Tendency (Sameness)

Plots and histograms show where values fall, but lack precision. We can summarize data with a single number representing its "center" - these **measures of central tendency** show what values are generally like.

2.3.1 Mode

The **mode** is the most frequently occurring value in a dataset. How do you find it? You have to count the number of times each number appears, then whichever one occurs the most, is the mode.

Example: 1 1 1 2 3 4 5 6

The mode is 1, since it appears three times.

If two or more values tie for most frequent, the dataset has multiple modes:

> Example: 1 1 1 2 2 2 3 4 5 6

Here, 1 and 2 both occur three times each. So, there are two modes, and they are 1 and 2.

Is the mode a good measure of central tendency? That depends on your numbers. The mode can be useful when a dataset has clear repeating values, but it doesn't always capture what's "typical." For example:

> 1 1 2 3 4 5 6 7 8 9

Here, the mode is 1, but most numbers are larger. Like any statistical tool, consider if the mode suits your dataset and justify that choice.

2.3.2 Median

The **median** is the exact middle of the data once they are ordered from smallest to largest. For example:

> 1 5 4 3 6 7 9

Before we can compute the median, we need to order the numbers from smallest to largest. Ordered:

> 1 3 4 **5** 6 7 9

The median is 5, with three numbers on each side.

With an even number of values, the median is the midpoint between the two in the center:

> 1 2 3 4 5 6

Here we have six numbers, so there isn't a single middle value. Instead, the median is the midpoint between the two central numbers, 3 and 4, which gives 3.5.

The median is often a useful measure of central tendency because it stays put even if some values are extreme. For example:

> 1 2 3 4 4 4 **5** 6 6 6 7 7 1000

Most of these numbers are modest, but 1000 is far from the rest. The median is still 5, which reflects the bulk of the data despite that extreme value. This shows why the median often does a good job of representing a dataset even when one or two values are very different.

In this case, 1000 would be considered an **outlier**: a value much farther from the others. Outliers can strongly affect some summaries (like the mean) but leave the median unchanged. How to handle outliers is a topic we return to later in the course.

2.3.3 Mean

Have you noticed this statistics textbook hasn't used a formula yet? That's about to change, but don't worry if you have formula anxiety - we'll explain them. The **mean** is also called the average. You probably know it's the sum of numbers divided by how many numbers there are.

Here's the formula:

$Mean = \bar{X} = \frac{\sum_{i=1}^{n} x_i}{N}$

- The $\sum$ symbol is called **sigma**, and it stands for the operation of summing.
- The "i" and "n" refer to all numbers in the set, from first to last.
- The x_i refers to individual numbers in the set.
- $\bar{X}$ refers to the mean
- We sum them all, then divide by N, the total count.

In simpler terms:

$mean = \frac{\text{Sum of my numbers}}{\text{Count of my numbers}}$

Let's compute the mean for these five numbers:

3 7 9 2 6

Add them:

3+7+9+2+6 = 27

Count them:

$i_1 = 3$, $i_2 = 7$, $i_3 = 9$, $i_4 = 2$, $i_5 = 6$; N=5, because i went from 1 to 5

Divide:

mean = 27 / 5 = 5.4

Or, putting everything in the formula:

$Mean = \bar{X} = \frac{\sum_{i=1}^{n} x_i}{N} = \frac{3+7+9+2+6}{5} = \frac{27}{5} = 5.4$

That's how to compute the mean. You probably knew this already, and if not, now you do. But is the mean a good measure of central tendency? By now, you should know: it depends.

2.3.4 What does the mean mean?

It's not enough to know how to calculate a mean, you also need to understand what it represents. The formula divides the sum of all values by the number of values, but what does that operation actually do?

Think about division. If we compute

$\frac{12}{3} = 4$

we're splitting 12 into three equal parts. Each part is 4. Division equalizes the numerator into identical pieces.

"The same logic applies to the mean. Suppose you add up all 500 happiness ratings into one large total. If that total were redistributed equally across every person, each would get the same value. That value is the mean. Some individuals were originally higher or lower, but the mean is the balance point created by spreading the total evenly.

This is why the mean is more than just arithmetic. It is the one number that can replace every observation so that, when all those replacements are added together, you get back the original total.

> **i** Takeaway
>
> The mean is unique: it is the only single value that can replace every observation so that the total stays the same.

2.3.5 Comparing the mean, median, and mode

Figure **??** shows a histogram of simulated data with three vertical lines: the mode (blue, 1), the median (green, 6), and the mean (red, 8).

Notice that these three measures of central tendency do not agree. The **mean** is pulled to the right because of a few very large values in the tail. The **median** is more stable, landing near the middle of the bulk of the data. The **mode** marks the most common single value, which here is close to zero.

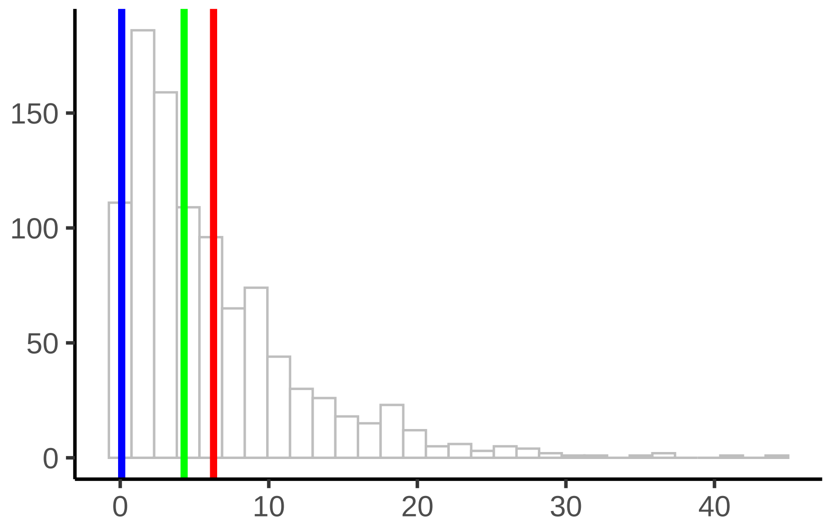

Figure 2.4: A histogram with the mean (red), the median (green), and the mode (blue)

> **Takeaway**
>
> When data are skewed—as they are here with a long right tail—the mean, median, and mode can give very different answers. Understanding how each behaves is essential for choosing the right summary.

2.4 Measures of Variation (Different*ness*)

Central tendency tells us what values have in common. Measures of **variation** tell us how they differ. Any dataset with more than one value will show some variation, and summarizing that spread is as important as finding the center.

2.4.1 The Range

Consider these 10 ordered numbers:

> 1 3 4 5 5 6 7 8 9 24

The smallest is 1, the largest is 24. Together they define the **range**. The range quickly shows the boundaries of the data and can flag possible outliers. For instance, if you expected values between 1 and 7 but found one at 340,500, you'd know something unusual happened and might investigate.

2.4.2 Difference Scores

It would be nice to summarize the amount of different*ness* in the data. Think about these 10 ordered numbers:

1 3 4 5 5 6 7 8 9 24

We can compute the differences between each pair of numbers and put them in a matrix:

	1	3	4	5	5	6	7	8	9	24
1	0	2	3	4	4	5	6	7	8	23
3	-2	0	1	2	2	3	4	5	6	21
4	-3	-1	0	1	1	2	3	4	5	20
5	-4	-2	-1	0	0	1	2	3	4	19
5	-4	-2	-1	0	0	1	2	3	4	19
6	-5	-3	-2	-1	-1	0	1	2	3	18
7	-6	-4	-3	-2	-2	-1	0	1	2	17
8	-7	-5	-4	-3	-3	-2	-1	0	1	16
9	-8	-6	-5	-4	-4	-3	-2	-1	0	15
24	-23	-21	-20	-19	-19	-18	-17	-16	-15	0

In the top left, the difference between 1 and itself is 0. One column over, 3–1 = 2, and so on. With 10 numbers this produces 100 differences. With 500 numbers, you'd have 250,000—too many to be useful.

If all the numbers were the same, every difference would be 0, making the lack of variation obvious. But with different numbers, we end up with a large table. How can we summarize it? One idea is to apply what we learned about central tendency and take the **average difference**.

Let's try it with three numbers:

1 2 3

	1	2	3
1	0	1	2
2	-1	0	1
3	-2	-1	0

The mean of these nine difference scores is:

mean of difference scores $= \frac{0+1+2-1+0+1-2-1+0}{9} = \frac{0}{9} = 0$

This will always happen: the positives and negatives cancel. So the mean of raw difference scores is not a useful measure of variation.

Notice also that the matrix is redundant: the diagonal is always zero, and values above and below the diagonal are the same except for sign.

These problems motivate why we compute **variance** and **standard deviation**. They solve the cancellation issue and give us a practical summary of spread.

2.4.3 The Variance

We've used the words variability, variation, and variance a lot. They all point to the same big idea: numbers differ. When numbers are different, they have variance.]

The word *variance* is used in two ways. First, it can mean the general idea of differences between numbers—when values vary, there is variance. Second, it refers to a specific summary statistic: the **mean of the squared deviations from the mean**.

To calculate it, we take each score, subtract the mean to find its **difference score**, then square those differences, then average them. Squaring is the mathematical "trick" that keeps positive and negative deviations from canceling each other out. Finally, we average the squared deviations. In short:

$variance = \frac{\text{Sum of squared difference scores}}{\text{Number of Scores}}$

Later we'll distinguish between dividing by N (all data = population) or $N - 1$ (sample). For now, just use N.

2.4.3.1 Deviations from the mean

Earlier we compared every number to every other number, which quickly became unmanageable. A simpler approach is to compare each score to the mean.

- Step 1: Find the mean.
- Step 2: Subtract the mean from each score.

This tells us:

1. How well the mean represents the data
2. How much spread there is around that mean.

Here's an example:

scores	values	mean	Difference_from_Mean
1	1	4.5	-3.5
2	6	4.5	1.5
3	4	4.5	-0.5
4	2	4.5	-2.5
5	6	4.5	1.5
6	8	4.5	3.5
Sums	27	27	0
Means	4.5	4.5	0

The mean is 4.5:

$\frac{1+6+4+2+6+8}{6} = \frac{27}{6} = 4.5.$

The third column simply repeats this mean for each row. That looks odd, but it shows the important property I mentioned earlier: the mean distributes the total equally across all points. Six copies of 4.5 add back to the total of 27.

The fourth column shows the **deviation from the mean:** $X_i - \bar{X}$. For example, 1, is -3.5 from the mean, 6, is +1.5, and so on.

Now notice the problem: the deviations add up to zero. The positive and negative differences cancel, which makes it seem like there's no variation. Clearly that isn't true, so we'll need another step, squaring the deviations, to solve it. But first let's investigate *why* this happens.

2.4.3.2 Mean as the Balancing Point

The mean is the balancing point of the data. Imagine laying a ruler across your finger. The place where it balances is the point where the weight on each side is equal. Data works the same way: if we treat each value like a weight, the mean is where the total "mass" to the left and right cancel out.

This balancing property explains why the deviations always sum to zero. The values below the mean contribute negative deviations, the values above contribute positive ones, and together they cancel:

$-x + x = 0$

To see this more concretely, consider the numbers

$X = (1, 2, 6, 7, 9)$

Figure 2.5: Values on a number line.

Now imagine the number line as a teeter-totter. Only when the fulcrum is placed at 5 does the board balance:

Formally, this means the signed distances from the mean always cancel out:

$(1-5)+(2-5)+(6-5)+(7-5)+(9-5)=0$

This makes the mean unique. It is the **only value** for which the sum of deviations equals zero:

$\sum_{i=1}^{n} x_{i-a} = 0 \quad \Rightarrow \quad a = \bar{x}$

The mean is not just a "typical" value, it is the one point where the data balance. And that balancing property is exactly why the raw deviations always sum to zero. To summarize variation, we need a way around this problem.

2.4.3.3 The squared deviations

The standard trick is to square the deviations. Squaring converts negatives to positives:

$2^2 = 4$

$-2^2 = 4.$

Since a squared number is always non-negative, the deviations no longer cancel. We call these **squared deviations**: differences from the mean that have been squared.

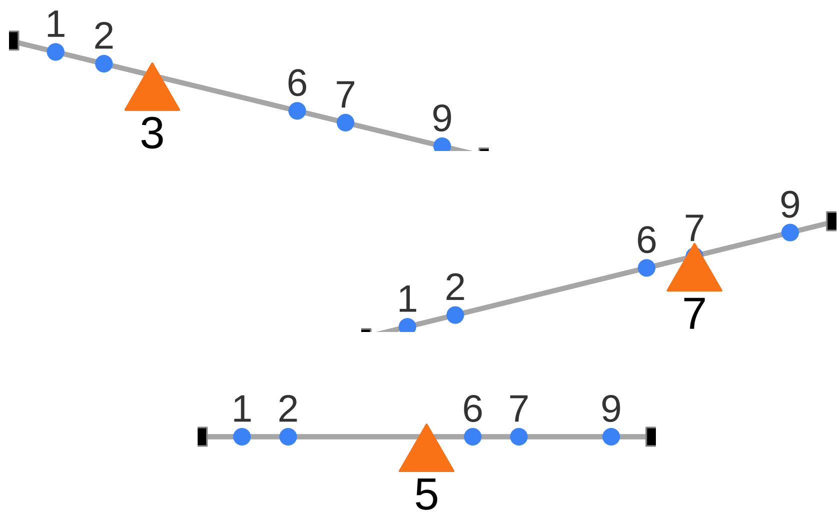

Figure 2.6: Mean as the unique balancing point. Only at the mean do signed deviations cancel.

Let's revisit our table, this time adding a column for squared deviations:

scores	values	mean	Difference_from_Mean	Squared_Deviations
1	1	4.5	-3.5	12.25
2	6	4.5	1.5	2.25
3	4	4.5	-0.5	0.25
4	2	4.5	-2.5	6.25
5	6	4.5	1.5	2.25
6	8	4.5	3.5	12.25
Sums	27	27	0	35.5
Means	4.5	4.5	0	5.91666666666667

For example, the first score is 1, which is 3.5 below the mean. Its deviation is -3.5, and its squared deviation is $(-3.5)^2 = 12.25$.

Now that all deviations are positive, we can add them up. The result is the **sum of squares (SS)**, the sum of squared deviations from the mean. You'll see this quantity again in the ANOVA chapter, but the idea is simple: it's just the total of those squared deviations, nothing more.

2.4.3.4 Finally, the variance

Guess what, we've already computed the variance. Maybe you didn't notice.

Let's remind ourselves of the goal: we want a single number that summarizes how spread out the data are. Deviations from the mean show those differences, but there are as many deviations as data points. To simplify, we want the *average* squared deviation.

Look back at the table. We added up the squared deviations (the *sum of squares*, SS), and then we divided by the number of observations. That's **the variance**. The variance is the mean of the sum of the squared deviations:

$variance = \frac{SS}{N}$

where SS is the sum of squared deviations, and N is the number of observations.

For our data, this came out to 5.916 (repeating).

So what does that number mean? Honestly, not much on its own. The problem is that it's on the *squared* scale, remember, we squared the deviations before averaging. Squaring inflates the values, so the variance is no longer in the same units as the original data.

The fix is straightforward: take the square root. For our example,

$\sqrt{5.916} \approx 2.43$

2.4.4 The Standard Deviation

We did it again, we already computed the **standard deviation (SD)**, without calling it by name. The standard deviation is just the square root of the variance:

standard deviation $= \sqrt{Variance} = \sqrt{\frac{SS}{N}}$.

or, written out fully,

standard deviation $= \sqrt{\frac{\sum_i^n (x_i - \bar{x})^2}{N}}$

Those square root signs simply "unsquare" the variance, bringing the measure of spread back to the original scale of the data. Let's look at our table again:

scores	values	mean	Difference_from_Mean	Squared_Deviations
1	1	4.5	-3.5	12.25
2	6	4.5	1.5	2.25
3	4	4.5	-0.5	0.25
4	2	4.5	-2.5	6.25
5	6	4.5	1.5	2.25
6	8	4.5	3.5	12.25
Sums	27	27	0	35.5
Means	4.5	4.5	0	5.91666666666667

We calculated our standard deviation as:

$\sqrt{5.916} \approx 2.43$

This value makes much more sense than the variance. A variance of 5.916 feels too large for this dataset, because it's on the squared scale. But a standard deviation of 2.43 lines up with the actual differences from the mean—most scores are within about ±2.4 of 4.5.

So if someone told you their dataset had a mean of 4.5 and a standard deviation of 2.4, you'd already have a good sense of it: the numbers cluster around 4.5, but not exactly at 4.5, and the typical spread is about 2 to 3 units in either direction.

That's the power of the standard deviation: it gives you a single number that describes the *typical deviation* from the mean, while staying on the same scale as the original data.

2.4.5 Mean Absolute Deviation

So far we've handled the "differences cancel out" problem by squaring deviations. But there's another simple option: take the **absolute value** of each deviation from the mean. That way, every difference becomes positive, and when we add them up we don't end up at zero.

Here's what that looks like for our data:

scores	values	mean	Difference_from_Mean	Absolute_Deviations
1	1	4.5	-3.5	3.5
2	6	4.5	1.5	1.5
3	4	4.5	-0.5	0.5
4	2	4.5	-2.5	2.5
5	6	4.5	1.5	1.5
6	8	4.5	3.5	3.5
Sums	27	27	0	13
Means	4.5	4.5	0	2.16666666666667

The last column shows absolute deviations. If we take their mean, we get the mean absolute deviation (MAD). Notice this acts a lot like the standard deviation: it's the average size of a deviation from the mean, only without squaring.Both approaches—squaring or taking absolute values—solve the same problem in slightly different ways. Squaring is more common because it leads to nice mathematical properties (like in regression and ANOVA), but the MAD can be more intuitive to interpret.

2.5 Remember to look at your data

Descriptive statistics are useful, but they are also compressed summaries. They reduce a dataset to a few numbers, which means they always lose detail. That's fine if the summary captures the important features, but sometimes it doesn't.

The safest habit is to **always combine descriptive statistics with graphs**. Graphs show you the shape of the data and reveal patterns summaries can hide.

2.5.1 Anscombe's Quartet

Francis Anscombe (Anscombe (1973)) made this point with a famous example, now called *Anscombe's Quartet*. Each panel in Figure Figure **??** shows pairs of x and y values as a scatterplot.

Visually, the four panels are very different: one looks linear, one curved, one is mostly linear except for an outlier, and one forms a near-vertical line except for one point.

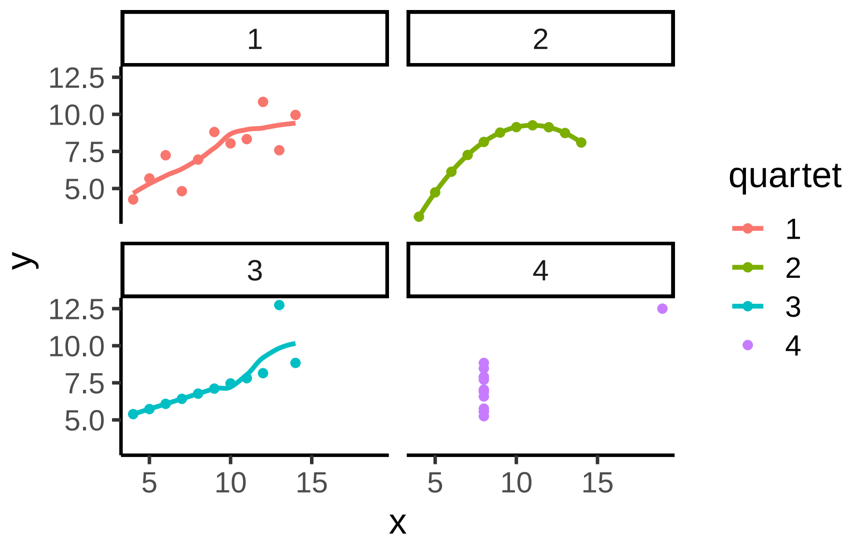

Figure 2.7: Anscombe's Quartet

Now here's the kicker:

	quartet	mean_x	var_x	mean_y	var_y
1		9	11	7.500909	4.127269
2		9	11	7.500909	4.127629
3		9	11	7.500000	4.122620

	quartet	mean_x	var_x	mean_y	var_y
4		9	11	7.500909	4.123249

All four datasets have the **exact same descriptive statistics**:

same mean and variance for x

same mean and variance for y

same correlation between x and y

Yet the scatterplots could not look more different.

> **Takeaway**
>
> Descriptive statistics alone can be misleading. Always look at the graph. Numbers may be identical, but patterns can be radically different.

2.5.2 Datasaurus Dozen

If you thought that Anscombe's quartet was neat, you should take a look at the Datasaurus Dozen (Matejka and Fitzmaurice 2017). hey constructed 13 datasets with nearly identical means, variances, and correlations — but when plotted, the points form shapes like a star, a circle, or even a dinosaur. Another reminder: your numbers might look like a dinosaur if you don't plot them.

2.6 Videos

2.6.1 Measures of center: Mode

2.6.2 Measures of center: Median and Mean

2.6.3 Standard deviation part I

2.6.4 Standard deviation part II

3 Correlation

> Correlation does not equal causation .*
> – Every Statistics Instructor Ever

In the last chapter, we worked with data that felt overwhelming at first. We used plots and histograms to make the data visible, and descriptive statistics to summarize their central tendency and variability.

The reason we learned those tools is simple: we want to use data to answer questions. That theme runs through this course. As you read each section, keep asking yourself: how does this concept help me answer questions with data?

In Chapter 2, we imagined collecting self-reports of happiness from a group of people. The data were just numbers, but they already raised useful questions. How spread out are people's happiness ratings? Do most people cluster near the middle, or are there groups of especially happy or unhappy individuals?

We also saw that numbers are imperfect reflections of experience. A rating from 0 to 100 is just one way of measuring happiness. It may not capture the full construct, but for now we will treat it as meaningful enough to work with.

This sets up the next step: moving from description to explanation. Rather than just asking how much happiness people report, we might ask what predicts differences in happiness? For example, does happiness vary with weather, income, or education? With social connections or cultural factors? Or with something lighter, like access to chocolate? Questions like these push us beyond describing data toward identifying relationships — the core of correlation analysis.

3.1 If something caused something else to change, what would that look like?

To use data for explanation, we need tools for recognizing when two things move together. Sometimes that reflects a real cause-and-effect relationship, and sometimes it doesn't. The challenge is learning to tell the difference. In this chapter we'll build intuition for what data look like when two variables are unrelated, when they move together, and when the pattern might mislead us.

3.1.1 More Chocolate = More Happiness?

Let's imagine that a person's supply of chocolate has a causal influence on their level of happiness. Let's further imagine that the more chocolate you have the more happy you will be, and the less chocolate you have, the less happy you will be. Finally, because we suspect happiness is caused by lots of other things in a person's life, we anticipate that the relationship between chocolate supply and happiness won't be perfect. What do these assumptions mean for how the data should look?

Our first step is to collect some imaginary data from 100 people. We walk around and ask the first 100 people we meet to answer two questions:

1. how much chocolate do you have? and
2. how happy are you?

For convenience, both the scales will go from 0 to 100. For the chocolate scale, 0 means no chocolate, 100 means lifetime supply of chocolate. Any other number is somewhere in between. For the happiness scale, 0 means no happiness, 100 means all of the happiness, and in between means some amount in between.

Here is some sample data from the first 10 imaginary subjects.

subject	chocolate	happiness
1	1	1
2	2	1
3	2	2
4	3	3
5	4	4
6	5	4
7	4	5
8	7	7
9	5	6
10	6	5

We asked each subject two questions so there are two scores for each subject, one for their chocolate supply, and one for their level of happiness. You might already notice some relationships between amount of chocolate and level of happiness in the table. To make those relationships even more clear, let's plot all of the data in a graph.

3.1.2 Scatter plots

When you have two measurements worth of data, you can always turn them into dots and plot them in a scatter plot. A scatter plot has a horizontal x-axis, and a vertical y-axis. You get to choose which measurement goes on which axis. Let's put chocolate supply on the x-axis, and happiness level on the y-axis. Figure **??** shows 100 dots. Each dot is for one subject and represents both their happiness level and their chocolate supply.

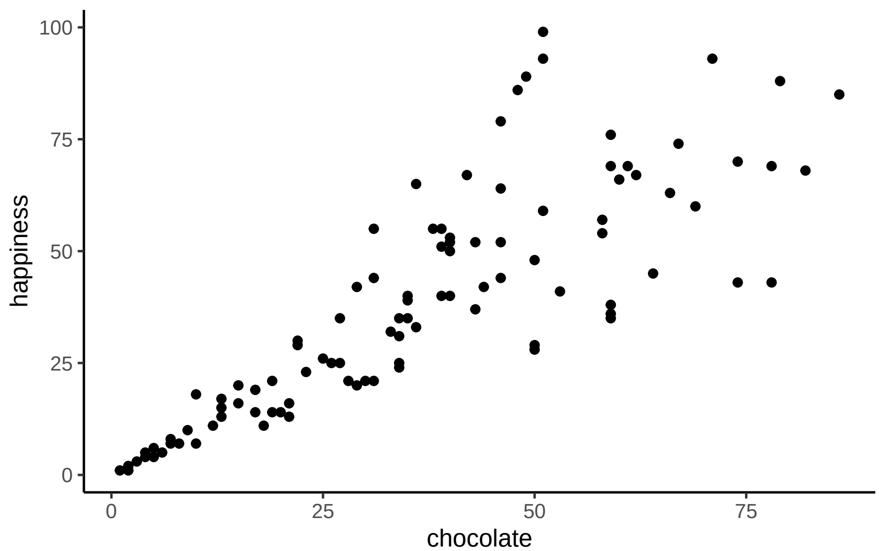

Figure 3.1: Imaginary data showing a positive correlation between amount of chocolate and amount happiness

Each dot has two coordinates, an x-coordinate for chocolate, and a y-coordinate for happiness. Looking at the **scatter plot**, we can see that the dots show a pattern. People with less chocolate tend to report lower happiness, while those with more chocolate tend to report higher happiness. It looks like the more chocolate you have the happier you will be, and vice-versa. This kind of relationship is called a **positive correlation**.

3.1.3 Positive, Negative, and No-Correlation

Let's imagine some more data. What do you imagine the scatter plot would look like if the relationship was reversed, and larger chocolate supplies decreased happiness? Or, what if there's no relationship, and the amount of chocolate that you have doesn't do anything to your happiness? take a look at Figure **??**:

The first panel shows a **negative correlation**. Happiness goes down as chocolate supply increases. Negative correlation occurs when one thing goes up and the other thing goes down;

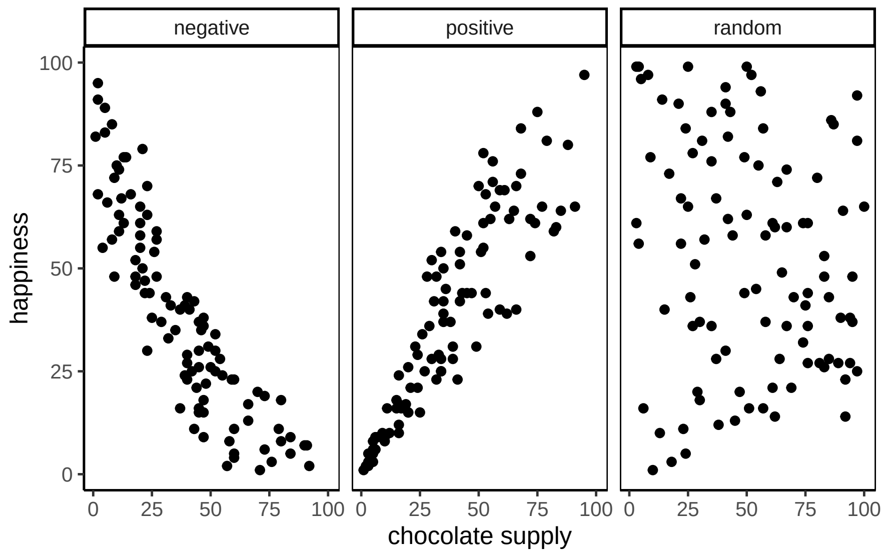

Figure 3.2: Three scatterplots showing negative, positive, and zero correlation

or, when more of X is less of Y, and vice-versa. The second panel shows a **positive correlation**. Happiness goes up as chocolate as chocolate supply increases. Positive correlation occurs when both things go up together, and go down together: more of X is more of Y, and vice-versa. The third panel shows **no correlation**. Here, there doesn't appear to be any obvious relationship between chocolate supply and happiness.

> Note
>
> **Always plot first.** If the scatter shows a U-shape or other curve, a single straight-line correlation will mislead. Fit a curved model (e.g., add a quadratic term) or transform variables instead of reporting r alone.

Zero correlation occurs when one thing is not related in any way to another things: changes in X do not relate to any changes in Y, and vice-versa.

You've examined your scatter plots, and now you might be wondering how to quantify what you see. We've already covered how to generate descriptive statistics for individual variables through means, standard deviations, etc. But what if you want to summarize how two variables move together in a single descriptive statistic? The path to that statistic (Pearson's r) starts with understanding covariance.

3.2 Covariance

Let's revisit **variance**. For one variable, variance summarizes how much values differ from their mean. **Covariance** extends this to two variables: it tells us whether being above the mean on X tends to come with being above the mean on Y (positive), below the mean on Y (negative), or neither (near zero).

In our chocolate–happiness example, the scatter plot suggested that higher chocolate tends to come with higher happiness. That means the deviations from each mean often have the same sign, so their products are mostly positive: evidence of *positive* covariance. Put simply, the two variables **vary together**.

Covariance is important because it underlies correlation, regression, and many other statistical tools. It can feel abstract at first. Keep the core idea in mind: look at how each variable's deviations from its mean behave together. If both are above or below their means at the same time, the covariance is positive. If one is above while the other is below, it's negative. If there's no consistent pairing, the covariance is near zero. With practice, this way of thinking becomes intuitive.

> **Note**
>
> **Covariance intuition.** Think of a three-legged race. When partners move forward together, their steps align (positive covariance). When one moves forward as the other moves back, their steps oppose (negative covariance). If their steps are uncoordinated, there's no consistent pattern (covariance near zero).

3.2.1 Turning the numbers into a measure of Covariance

If Covariance is just another word for correlation or relationship between two measures,then, we need some way to measure that. Consider our chocolate and happiness scores.

subject	chocolate	happiness	Chocolate_X_Happiness
1	1	1	1
2	2	1	2
3	2	2	4
4	3	3	9
5	4	4	16
6	5	4	20
7	4	5	20
8	7	7	49
9	5	6	30
10	6	5	30

subject	chocolate	happiness	Chocolate_X_Happiness
Sums	39	38	181
Means	3.9	3.8	18.1

We've added a new column called `Chocolate_X_Happiness`, which is simply each person's chocolate score multiplied by their happiness score. Why multiply? Because the products reveal how the two values move together.

- Small $\times$ small $\rightarrow$ small product
- Large $\times$ large $\rightarrow$ large product
- Small $\times$ large $\rightarrow$ product in between

Now extend this idea from pairs to the whole dataset. If chocolate and happiness rise together (both above average at the same time), the summed products will be large. If one tends to rise while the other falls, the summed products will shrink. This is the basic logic behind **covariance**.

Suppose we have two variables, X and Y. If X and Y go up together (both small at the same time, both large at the same time), their products will also be large when summed across the dataset. If one goes up while the other goes down (large pairs with small), the summed products will be much smaller.

Here's are two tables to illustrate:

Table 3.3: Aligned: sum = 385

X	Y	XY
1	1	1
2	2	4
3	3	9
4	4	16
5	5	25
6	6	36
7	7	49
8	8	64
9	9	81
10	10	100
55	55	385

Table 3.4: Reversed: sum = 220

A	B	AB
1	10	10
2	9	18
3	8	24
4	7	28
5	6	30
6	5	30
7	4	28
8	3	24
9	2	18
10	1	10
55	55	220

X and Y are perfectly aligned: 1 with 1, 2 with 2, up to 10 with 10. Their products add up to **385**, the *largest* possible sum.

A and B are perfectly reversed: 1 with 10, 2 with 9, and so on. Their products add up to **220**, the *smallest* possible sum.

Any other pairing of 1–10 with 1–10 will fall between these two extremes. Figure **??** shows the sums of products from 1000 random pairings. Every result falls between 220 (perfect negative alignment) and 385 (perfect positive alignment).

So what does this mean? The **sum of the products** tells us about the direction of the relationship:

- Close to 385 $\rightarrow$ strong positive relationship.
- Close to 220 $\rightarrow$ strong negative relationship.
- Near the middle $\rightarrow$ little or no relationship.

This is the foundation of *covariance*: products of paired values reveal whether two variables rise together, fall together, or move in opposite directions.

Takeaway:

- Positive correlation $\rightarrow$ large values in X pair with large values in Y $\rightarrow$ sum of products near the maximum.
- Negative correlation $\rightarrow$ large values in X pair with small values in Y $\rightarrow$ sum near the minimum.
- No correlation $\rightarrow$ pairings are random $\rightarrow$ sum lands in the middle.

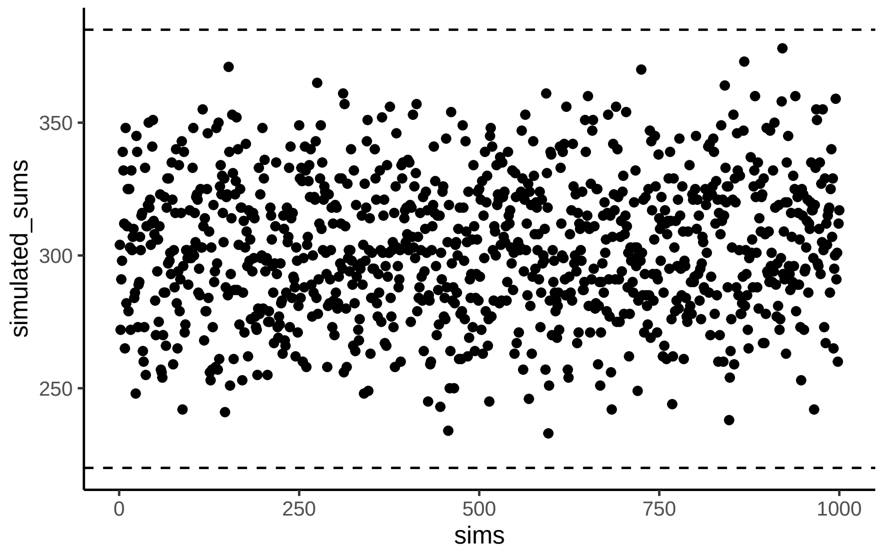

Figure 3.3: Simulated sums of products showing the kinds of values than can be produced by randomly ordering the numbers in X and Y.

3.2.2 Covariance, the measure

Earlier we saw that multiplying values from two variables can highlight how they vary together: large-with-large and small-with-small pairs push the product up, while mixed pairs keep it lower. That intuition is useful, but in statistics we define **covariance** more precisely.

Instead of multiplying raw values of X and Y, we multiply their *deviations from the mean*. This centers each variable, so the product reflects how the two variables co-vary around their averages:

$cov(X,Y) = \frac{\sum_i^n (x_i - \bar{X})(y_i - \bar{Y})}{N}$

In practice, the steps are straightforward: subtract the mean of X from each x_i, subtract the mean of Y from each y_i, multiply the deviations, and then average those products. A positive covariance means the two variables tend to move together (above or below their means at the same time). A negative covariance means they move in opposite directions. A covariance near zero suggests no consistent relationship.

The following table shows this process:

subject	chocolate	happiness	C_d	H_d	Cd_x_Hd
1	1	1	-2.9	-2.8	8.12
2	2	1	-1.9	-2.8	5.32
3	2	2	-1.9	-1.8	3.42

subject	chocolate	happiness	C_d	H_d	Cd_x_Hd
4	3	3	-0.9	-0.8	0.72
5	4	4	0.1	0.2	0.02
6	5	4	1.1	0.2	0.22
7	4	5	0.1	1.2	0.12
8	7	7	3.1	3.2	9.92
9	5	6	1.1	2.2	2.42
10	6	5	2.1	1.2	2.52
Sums	39	38	0	0	33
Means	3.9	3.8	0	0	3.28

Deviations from the mean for chocolate (column `C_d`), deviations from the mean for happiness (column `H_d`), and their products. The mean of those products (in the bottom right corner of the table) is the official **covariance**.

Covariance is a powerful idea, but it has a major drawback: its magnitude is hard to interpret. Just as variance is tied to squared units, covariance is tied to the scales of both variables. Is our covariance of 3.22 large? What if it's 100? The raw number doesn't tell us. All we can tell is its sign (positive or negative), not its size, which means we can't say how strong the correlation is. To compare strength across variables with different units or scales, we need to normalize covariance, which leads to Pearson's r.

3.2.3 Pearson's r

We often want a relationship measure that is **directional**, **unitless**, and **bounded**. **Pearson's** r does exactly this by rescaling covariance to lie between -1 and 1:

- $r = 1$: perfect positive linear relationship
- $r = 0$: no linear relationship
- $r = -1$: perfect negative linear relationship

Values closer to ± 1 indicate stronger linear association; values near 0 indicate weak or no linear association.

Let's take a look at a formula for Pearson's r:

$r = \frac{cov(X,Y)}{\sigma_X \sigma_Y} = \frac{cov(X,Y)}{SD_X SD_Y}$

In this forumal we see the symbol σ, which indicates the standard deviation (SD). In words, r is the Covariance of X and Y divided by the product of their standard deviations. Dividing by the standard deviations **normalizes** the Covariance, constraining r to fall between between -1 to 1. This makes correlation values comparable across variables measured in different units or on different scales.

Note

Computing *r*. You'll see algebraically equivalent formulas for Pearson's r. They produce the same result; we'll typically rely on software and focus on interpretation.

There are several equivalent formulas for computing Pearson's r. Some are written to simplify hand calculations, others are more compact algebraically. They all yield the same result. Since we rely on software for calculations, our focus will be on what r means and how to interpret it, rather than practicing manual computation. You'll see how to run the calculation in R during lab.

Does Pearson's r really stay between -1 and 1 no matter what? It's true, take a look at the following simulation. Here I randomly ordered the numbers 1 to 10 for an X measure, and did the same for a Y measure. Then, I computed Pearson's r, and repeated this process 1000 times. As you can see from Figure **??** all of the dots are between -1 and 1.

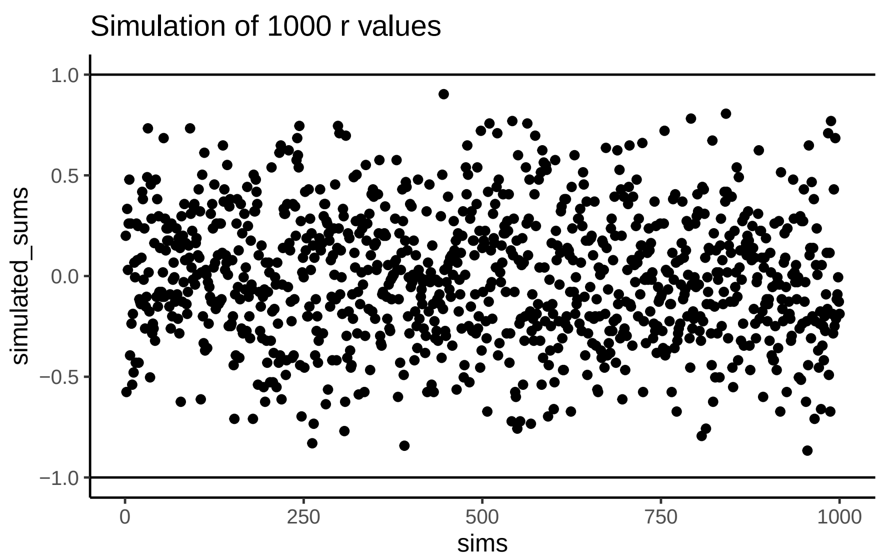

Figure 3.4: A simulation of of correlations. Each dot represents the r-value for the correlation between an X and Y variable that each contain the numbers 1 to 10 in random orders. The figure illustrates that many r-values can be obtained by this random process

3.3 Regression: A mini intro

We're going to spend the next little bit adding one more thing to our understanding of correlation. It's called **linear regression**. Just an introduction to the basic concepts.

First, let's look at a linear regression. This way we can see what we're trying to learn about. Figure **??** shows the same scatter plots as before with something new: lines!

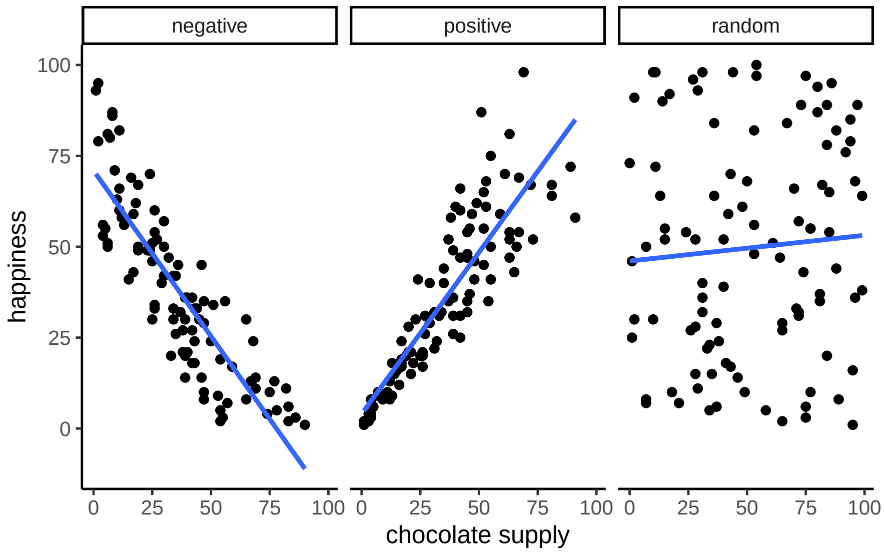

Figure 3.5: Three scatterplots showing negative, positive, and a random correlation (where the r-value is expected to be 0), along with the best fit regression line

3.3.1 The best fit line

Notice the blue lines in these panels. Each is a **best fit** line: the single straight line that best summarizes the overall pattern of dots.

With one variable, we use hte mean to describe central tendency in one dimension. With two variables plotted together, we need a two-dimensional equivalent. A line serves that purpose, capturing the average relationship across the could of points.

Of course, many possible lines could be drawn. Most would miss the pattern. The useful lines would go through the data following the general shape of the dots. Which one is the best? How can we find out, and what do we mean by that? In short, the best fit line is the one that has the least error.

Check out this plot: a line runs through the dots, and short vertical lines drop from each dot to the line. These are **residuals**:the vertical differences between the observed data and the fitted line. Residuals show the error, or how far off the line's prediction is for each point. Since not all dots fall exactly on the line, some error is inevitable. The best-fit line is the one that minimizes these errors overall.

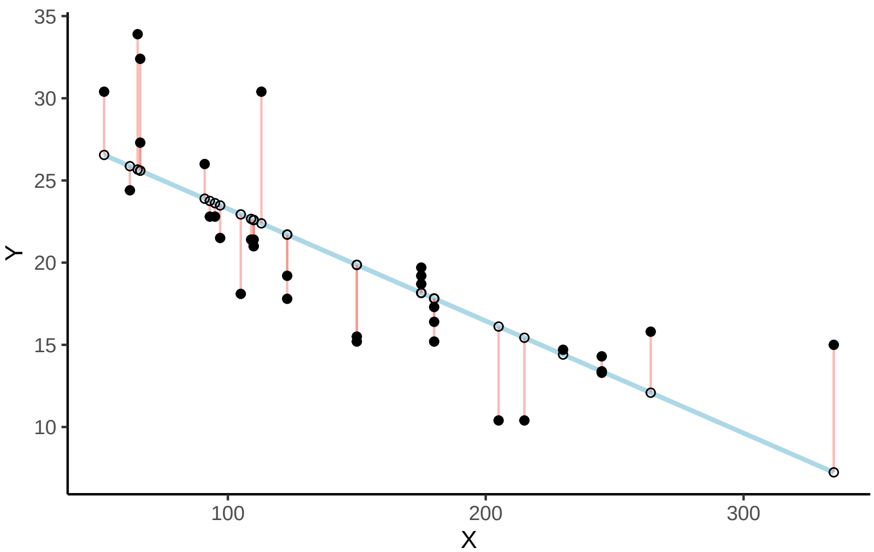

Figure 3.6: Black dots represent data points. The blue line is the best fit regression line. The white dots are repesent the predicted location of each black dot. The red lines show the error between each black dot and the regression line. The blue line is the best fit line because it minimizes the error shown by the red lines.R code for residual visualziation adapted from Simon Jackson: https://drsimonj.svbtle.com/visualising-residuals

There's a lot going on in Figure **??**. his is a scatter plot of two variables X and Y. Each black dot represents the observed data. The blue line is the regression line, and the white dots show the predicted values from that line. The red lines are **residuals**: they drop vertically from each observed value to its predicted value on the line. Because most of the black dots are not exactly on the line, each residual shows how far the prediction differs from the observed value.

The key point is that the regression line is chosen to minimize the total size of these residuals. In practice, we compute the **sum of squared deviations**: squaring each residual, then adding them together. The blue line is the one that produces the smallest possible total squared error. Any other line would yield a larger total error.

?@fig-3regressionGIF is an animation to see this in action. The animations compares the best fit line in blue, to some other possible lines in black. The black line moves up and down. The red lines show the error between the black line and the data points. As the black line moves toward the best fit line, the total error, depicted visually by the grey area shrinks to its minimum value. The total error expands as the black line moves away from the best fit line.

Whenever the black line does not overlap with the blue line, it fits the data less well. The blue regression line is the one that minimizes the overall error — not too high, not too low, but the line with the least total deviation.

Figure **??** shows how the sum of squared deviations (the sum of the squared lengths of the red lines) behaves as we move the line up and down. What's going on here is that we are computing a measure of the total error as the black line moves through the best fit line. This represents the sum of the squared deviations. In other words, we square the length of each red line from the above animation, then we add up all of the squared red lines, and get the total error (the total sum of the squared deviations). The graph below shows what the total error looks like as the black line approaches then moves away from the best fit line. Notice, the dots in this graph start high on the left side, then they swoop down to a minimum at the bottom middle of the graph. When they reach their minimum point, we have found a line that minimizes the total error. This is the best fit regression line.

OK, so we haven't talked about the y-intercept yet. But, what this graph shows us is how the total error behaves as we move the line up and down. The y-intercept here is the thing we change that makes our line move up and down. As you can see the dots go up when we move the line down from 0 to -5, and the dots go up when we move the line up from 0 to +5. The best line, that minimizes the error occurs right in the middle, when we don't move the blue regression line at all.

3.3.2 Computing the best fit line

How do we find the line that minimizes the sum of squared deviations? In principle, you could try many possible lines and compare their errors, but that's not efficient. Instead, we use

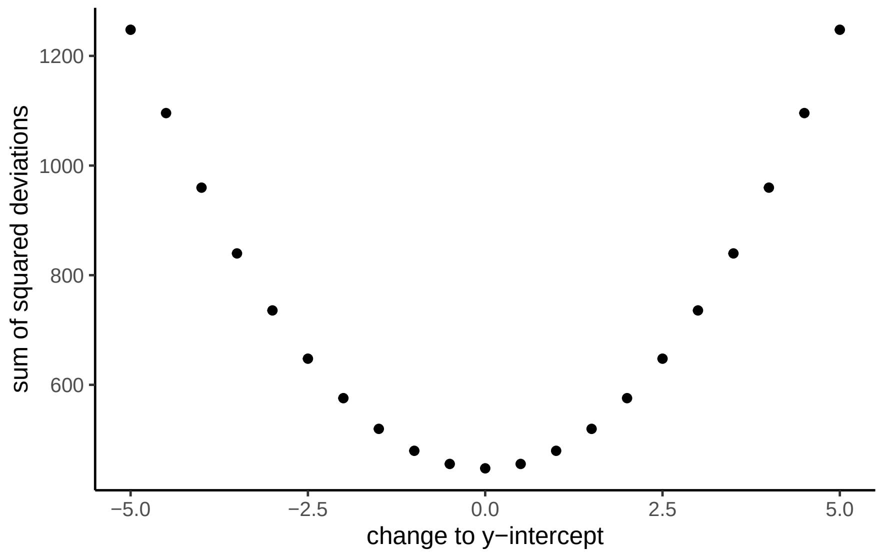

Figure 3.7: A plot of the sum of the squared deviations for different lines moving up and down, through the best fit line. The best fit line occurs at the position that minimizes the sum of the squared deviations.

formulas (or, more realistically, software) to identify the line with the smallest total error.

Next we'll look at the formulas for calculating the slope and intercept of a regression line, and work through one example by hand. Many statistical formulas were designed to simplify hand calculations before computers were common. Today we rely on software, but working through examples by hand helps illustrate the logic behind the formulas.

Here are two formulas we can use to calculate the slope and the intercept, straight from the data. We won't go into why these formulas do what they do. These ones are for "easy" calculation.

$intercept = b = \frac{\sum y \sum x^2 - \sum x \sum xy}{n \sum x^2 - (\sum x)^2}$

$slope = m = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$

In these formulas, the x and the y refer to the individual scores. Here's a table showing you how everything fits together.

	scores	x	y	x_squared	y_squared	xy
1		1	2	1	4	2
2		4	5	16	25	20
3		3	1	9	1	3
4		6	8	36	64	48

	scores	x	y	x_squared	y_squared	xy
5		5	6	25	36	30
6		7	8	49	64	56
7		8	9	64	81	72
Sums		34	39	200	275	231

We see 7 sets of scores for the x and y variable. We calculated x^2 by squaring each value of x, and putting it in a column. We calculated y^2 by squaring each value of y, and putting it in a column. Then we calculated xy, by multiplying each x score with each y score, and put that in a column. Then we added all the columns up, and put the sums at the bottom. These are all the number we need for the formulas to find the best fit line. Here's what the formulas look like when we put numbers in them:

$intercept = b = \frac{\sum y \sum x^2 - \sum x \sum xy}{n \sum x^2 - (\sum x)^2} = \frac{39*200-34*231}{7*200-34^2} = -.221$

$slope = m = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2} = \frac{7*231-34*39}{7*275-34^2} = 1.19$

Great, now we can check our work, let's plot the scores in a scatter plot and draw a line through it with slope = 1.19, and a y-intercept of -.221. As shown in Figure **??**, the line should go through the middle of the dots.

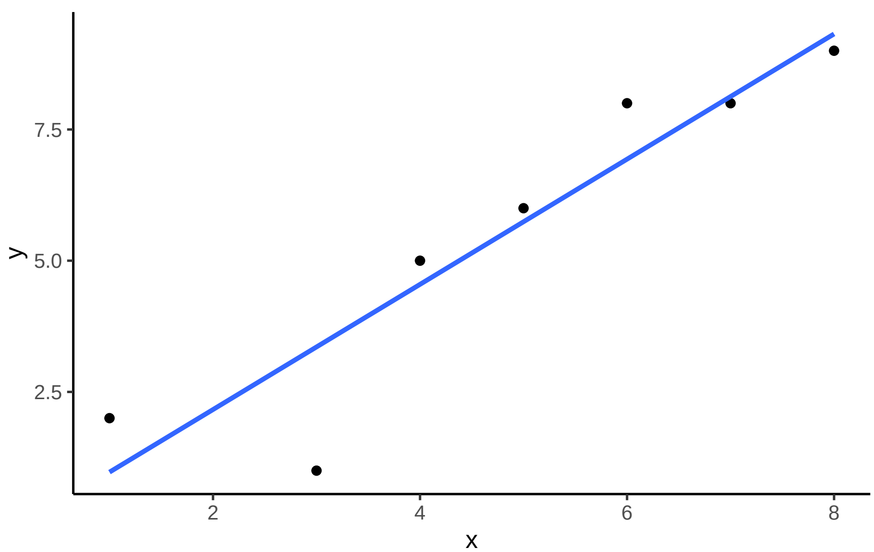

Figure 3.8: An example regression line with confidence bands going through a few data points in a scatterplot

3.4 Interpreting Correlations: why caution is warranted

You've heard the warning: **correlation does not equal causation**. But what does a correlation really tell us, and what can it mislead us about? Interpreting correlations requires care, because the same number can arise from very different situations: random chance, confounding influences, or nonlinear patterns. In this section, we'll look at why correlations can be unreliable and how to recognize their limits.

3.4.1 Spurious correlation: chance alone can look convincing

Another very important aspect of correlations is the fact that they can be produced by random chance. This means that you can find a positive or negative correlation between two measures, even when they have absolutely nothing to do with one another. You might have hoped to find zero correlation when two measures are totally unrelated to each other. Although this certainly happens, unrelated measures can accidentally produce **spurious** correlations, just by chance alone.

Let's demonstrate how correlations can appear by chance when there is no causal connection between two measures. Imagine two independent participants, one at the North Pole and one at the South Pole. Each has a lottery machine containing balls numbered 1 through 10. With replacement, each participant randomly draws 10 balls and records the numbers. Because the draws occur independently and half a world apart, there is no possible causal link between the two sets of numbers. The outcomes are determined by chance alone.

Here is what the numbers on each ball could look like for each participant:

Ball	North_pole	South_pole
1	2	10
2	3	1
3	7	3
4	6	6
5	5	6
6	6	5
7	3	10
8	8	6
9	8	9
10	7	7

In this one case, if we computed Pearson's r, we would find that r = -0.1318786. But, we already know that this value does not tell us anything about the relationship between the balls chosen in the north and south pole. We know that relationship should be completely random, because that is how we set up the game.

The better question here is to ask what can random chance do? For example, if we ran our game over and over again thousands of times, each time choosing new balls, and each time computing the correlation, what would we find?First, we will find fluctuation. The r value will sometimes be positive, sometimes be negative, sometimes be big and sometimes be small. Second, we will see what the fluctuation looks like. This will give us a window into the kinds of correlations that chance alone can produce. Let's see what happens.

3.4.1.1 Monte-carlo: what chance produces

We can use a computer to repeat the lottery game as many times as we like. This process, repeated random sampling to explore possible outcomes, is called a **Monte Carlo simulation**.

I've coded a loop to run the game 1,000 times. Each time, both the North Pole and South Pole participants generate 10 random numbers. We then compute the correlation between their two sets. The result is 1,000 different values of Pearson's r, each produced under conditions where we know no real relationship exists.

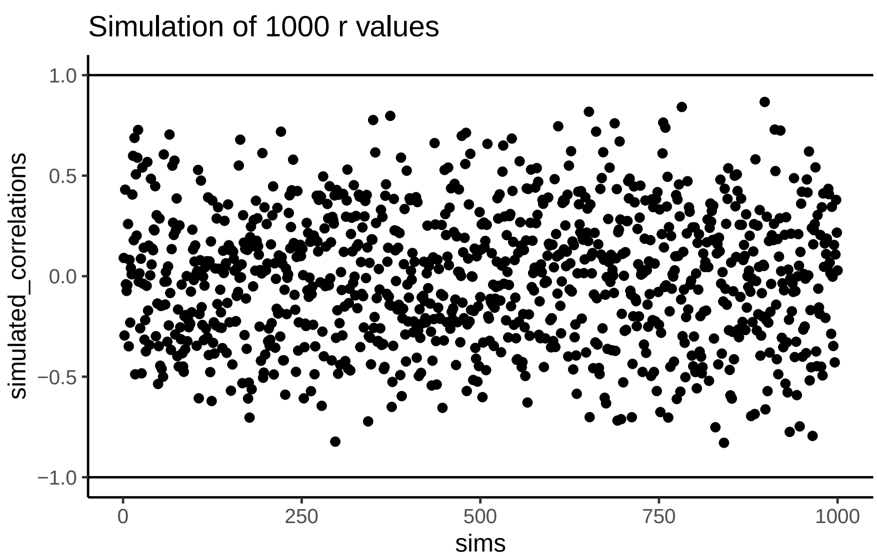

Figure 3.9: Another figure showing a range of r-values that can be obtained by chance.

Figure **??** shows the outcome. Each dot is one simulated correlation. All of them fall between –1 and 1, but the values scatter widely: sometimes strongly positive, sometimes strongly negative, often near zero. The point is that **chance alone can generate correlations of any size**. Finding a correlation in your data does not by itself guarantee a meaningful relationship.

To visualize this process more concretely, we can animate a single run. In **?@fig-3randcor10gif**, each frame shows two sets of 10 random values plotted against each other, along with the fitted regression line. Because the data are random, we expect no consistent pattern. Yet across repetitions, the line sometimes slopes upward, sometimes downward, and occasionally appears flat. This is what spurious correlation looks like.

The animation makes clear how quickly random samples can mimic each of these patterns.

You might find this unsettling: if two random variables with no relationship can still produce correlations, how can we trust any correlation at all? This raises a key question: **how do we know whether the correlation we observed is meaningful or just chance?** One way forward is to look not at a single outcome, but at the distribution of outcomes produced by chance. By simulating 1,000 random correlations and plotting them in a histogram, we can see which values occur frequently and which are rare.

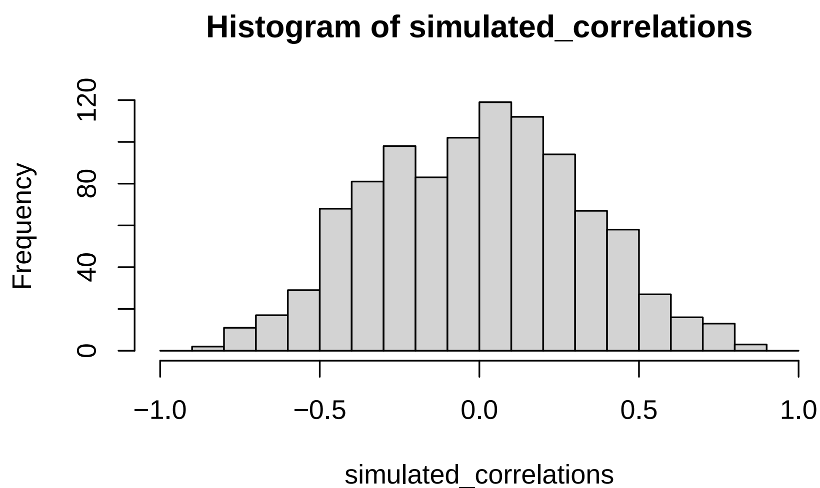

Figure 3.10: A histogram showing the frequency distribution of r-values for completely random values between an X and Y variable (sample-size=10). A rull range of r-values can be obtained by chance alone. Larger r-values are less common than smaller r-values

Figure **??** shows that most chance correlations cluster near zero. Moderate correlations (around ± 0.5) occur less frequently, and near-perfect correlations (± 0.95) are almost absent. This distribution defines a "window of chance." When we compute a correlation in real data, we can ask: is our observed r consistent with what chance alone could plausibly generate?

or example, if you observed $r = 0.1$, the histogram shows that such values are common in random data, so chance could easily explain it. An observed $r = 0.5$ is less common but still possible by chance. An observed $r = 0.95$, however, falls outside what chance typically produces, suggesting that something more systematic is driving the relationship.

3.4.2 Sample size and stability

So far, our lottery game has used 10 draws per participant. With such small samples, chance produces a wide range of correlations: sometimes strongly positive, sometimes strongly negative. But what happens if we increase the sample size?

We can repeat the simulation under four conditions: each participant draws 10, 50, 100, or 1000 numbers. In each case, we run 1000 simulations and record the correlation values.

Figure **??** shows the resulting four histograms of Pearson r values.

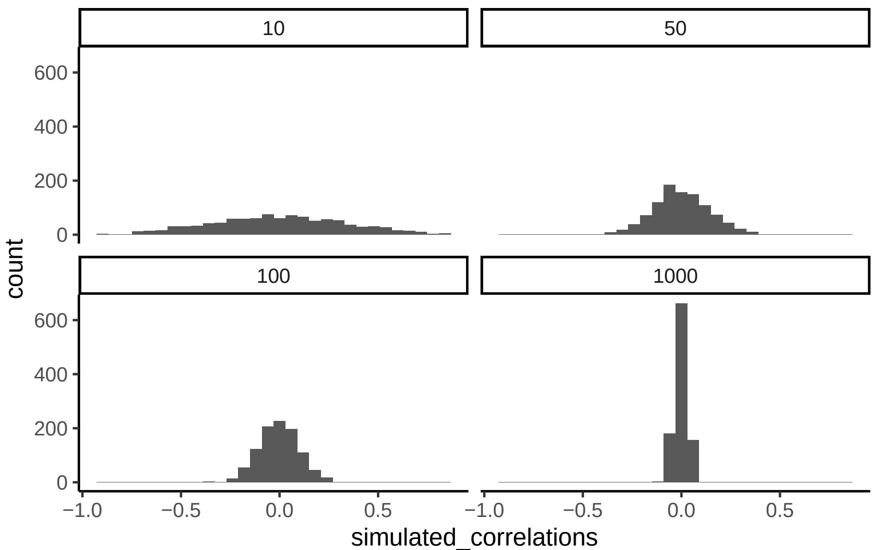

Figure 3.11: Four histograms showing the frequency distributions of r-values between completely random X and Y variables as a function of sample-size. The width of the distributions shrink as sample-size increases. Smaller sample-sizes are more likely to produce a wider range of r-values by chance. Larger sample-sizes always produce a narrow range of small r-values

The pattern is clear:

- With **n=10**, correlations spread widely across the range -1 to 1. Chance alone frequently produces values that would look "strong" in real data.
- With **n=50 or 100**, the spread narrows, but moderate correlations still appear.
- With **n=1000**, almost all simulated correlations fall close to zero.

The takeaway is that **larger samples reduce the influence of chance**. Small samples are highly unstable, making it difficult to distinguish signal from noise. As sample size increases, the distribution of chance correlations collapses toward zero, so observing a large r becomes far less likely under randomness alone.

This illustrates why researchers place so much emphasis on collecting sufficient data. With too few observations, even strong-looking correlations may be spurious. With larger samples, a correlation of the same size is much harder to attribute to chance, and therefore more likely to reflect a real underlying relationship.

> **Note**
>
> **Experimental design tip: sample size.** Bigger n narrows the "window of chance." If you expect a modest linear effect (e.g., |r| 0.2–0.3), plan for larger samples so random swings are less likely to mimic or mask the effect.

3.4.2.1 Visualizing the effect of sample size

Animations make it easier to see how sample size influences the stability of correlation estimates. When the sample is very small, chance variation can create apparent patterns in any direction. As the sample size increases, those chance fluctuations diminish.

When there is no true correlation

In **?@fig-3corRandfour**, each panel shows two random variables sampled with no real relationship. The panels differ only in sample size: 10, 50, 100, or 1000 observations. Each frame of the animation re-samples the data, plots them, and fits a line.

With $n = 10$, the line swings dramatically from positive to negative slopes. At $n = 50$ and $n = 100$, the line still shifts but is more restrained. At $n = 1000$, the line is consistently flat, as expected when no correlation exists.

Which line should you trust? The line from the sample with 1000 observations is the most reliable. It stays nearly flat across repetitions, as we would expect when no true correlation exists. The line from a sample of only 10 observations swings wildly, sometimes positive, sometimes negative. The lesson is straightforward: claims about correlation are only as trustworthy as the sample size behind them. Small samples can produce misleading patterns by chance, while large samples rarely do.

We can also repeat this simulation with data drawn from a normal distribution rather than a uniform one. The result is the same: small samples fluctuate dramatically, but with large samples the estimated correlation stays close to zero. **?@fig-3normCorfour** illustrates this.

What do things look like when there actually is a correlation between variables?

When there is a true correlation

In **?@fig-3realcorFour**, the simulation is repeated with a real positive correlation built into the data. Again, we vary the sample size from 10 to 1000.

With $n = 10$, the regression line jumps around and occasionally even points downward, despite the underlying positive relationship. As the sample size grows, the line becomes more stable and consistently reflects the true positive slope. Sampling error still introduces some noise, but with larger samples the real correlation is much easier to detect.

Both the histograms and the animations show that **small samples make correlations unreliable, while larger samples are more likely to reveal the true pattern**. Even when a real positive correlation exists, a small sample might miss it, or even suggest the opposite, simply due to chance. Because we usually only have one sample, we can never know for certain whether we were "lucky" or "unlucky." The best protection is to collect more data: larger samples reduce the influence of chance and make genuine relationships easier to detect.

3.4.3 Nonlinearity: when r misses real relationships

Consider, a snake plant. Like most plants, snake plants need water to stay alive. But, they also need *the right amount* of water. Imagine we grow 1000 snake plants, each receiving a different amount of water per day. The amount of water ranges from *no water* to *extreme overwatering* (say 1000 teaspoons of water per day). We then measure weekly snake plant growth. Water is clearly causally related to plant growth, so we expect to see a correlation.

What would the scatter plot look like?

Plants given no water at all will eventually die, showing little or no weekly growth. With only a few teaspoons per day, plants may survive but grow only modestly. In a scatter plot, those cases would appear near the bottom left (low water, low growth) and rise upward as water increases. Growth continues to improve until a threshold is reached—beyond that point, excess water harms the plants. Severely overwatered plants also fail to grow and eventually die, just as those given no water do.

The scatter plot would form an inverted U or V: as water increases, growth rises to a peak and then declines with overwatering. A relationship like this can produce a Pearson's r close to zero, even though water clearly affects growth. Figure **??** illustrates this pattern.

There is clearly a relationship between watering and snake plant growth. But, the correlation isn't in one direction. As a result, when we compute the correlation in terms of Pearson's r, we get a value very close to zero, suggesting no relationship.

```
#> [1] -0.02108469
```

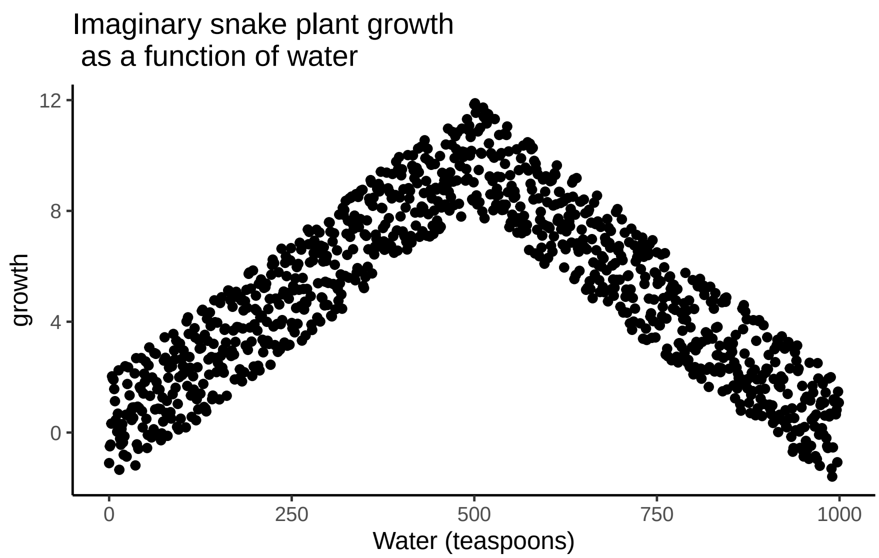

Figure 3.12: Illustration of a possible relationship between amount of water and snake plant growth. Growth goes up with water, but eventually goes back down as too much water makes snake plants die.

What this really means is there is no relationship that can be described by a *single straight line*. When we need lines or curves going in more than one direction, we have a nonlinear relationship.

This example shows why correlations can be tricky to interpret. Water is clearly necessary for plant growth, yet the overall correlation can be misleading. The first half of the data suggests a positive relationship, the second half a negative one, and the full dataset little or none. Even with a real causal link, a single correlation coefficient may fail to capture it.

> Pro Tip: This is one reason why plotting your data is so important. If you see an upside U shape pattern, then a correlation analysis is probably not the best analysis for your data.

3.4.4 Confounding variables (third-variable problem)

Correlations can arise because of a third variable—something not directly measured that influences both variables of interest. Returning to the snake plant example: imagine we measure water and plant growth, but ignore sunlight. If plants in sunnier windows both receive more water and grow faster, we might attribute the effect to water alone, when in fact sunlight is doing much of the work.

The same issue appears in human data. Suppose we find that income and happiness are positively correlated. Is money itself the cause? Possibly, but other factors – such as housing quality, healthcare access, or social opportunities – may be the true drivers. Income is correlated with these factors, which in turn influence happiness.

3.5 Summary

Why it matters. Correlation is how we quantify whether two variables move together. It's essential for describing patterns (e.g., temperature and canopy cover, chocolate and happiness) and for deciding whether a relationship is worth modeling or acting on.

Core ideas

- **Scatter first, always.** A scatter plot shows direction (up, down, none), form (straight vs curved), and outliers.
- **Covariance → correlation.** Covariance measures whether deviations from each mean tend to align. Pearson's r rescales covariance by the standard deviations so the result is unitless and bounded ([-1, 1]).
- **Interpreting *r*.** Sign indicates direction; magnitude indicates strength of a *linear* relationship. r near 0 can still hide a strong *nonlinear* pattern.
- **Regression link.** The best-fit line summarizes the average relationship; residuals are the vertical gaps between data and the line. The line is chosen to minimize the **sum of squared residuals**.

Common pitfalls

- **Chance (spurious) correlations.** Small samples can produce large $|r|$ by accident. Larger samples shrink the "chance window" toward 0.
- **Nonlinearity.** Curved relationships (e.g., too little or too much water harming plant growth) can yield r 0 despite a real effect.
- **Confounding.** A third variable can create or inflate a correlation (e.g., sunlight affecting both watering patterns and plant growth).

Practice checklist

1. Plot the scatter; look for linearness, curvature, and outliers.

2. Report r with **sample size** (n) and, when possible, a **confidence interval**.
3. Avoid causal language—say "associated with," not "caused by."

4. If the pattern is curved, don't stop at r; fit a suitable model (e.g., add a quadratic term) or transform.

5. If results hinge on a small sample, say so and treat conclusions as tentative.

Takeaway. Correlation is a precise way to summarize linear association, but it's only as trustworthy as your plot, your sample size, and your awareness of curvature and confounding. Plot first, quantify carefully, and interpret cautiously.

4 Probability and Distributions

> I have studied many languages-French, Spanish and a little Italian, but no one told me that Statistics was a foreign language. – Charmaine J. Forde

4.1 From description to inference

Up to this point, we've focused on experimental design and descriptive statistics: how to collect data and how to summarize it with graphs and averages. But statistics is more than description. Its power is in inference, using limited data to say something about a broader population. Inference relies on two foundations: probability theory and the behavior of samples from distributions.

4.2 Probability vs. Statistics

Probability and statistics are closely related but not the same.

Probability starts with a known model of the world and asks: what kinds of data are likely?

- Example: What is the chance of flipping 10 heads in a row with a fair coin?
- Example: What is the chance of drawing five hearts from a shuffled deck?

Here, the **model is known** (fair coin, fair deck). Data are imagined.

Statistics starts with data and asks: what model of the world generated these data?

- Example: If a coin comes up heads 10 times in a row, is it fair?
- Example: If five cards in a row are hearts, was the deck shuffled?

Here, the **data are known**. The model is what we want to learn about.

> Note
>
> Shortcut: - Probability = model → data. - Statistics = data → model.

4.3 What does probability mean?

Statisticians agree on the rules of probability but not always on what the word itself means. In everyday life we're comfortable saying something is "likely" or "unlikely," but the formal meaning varies.

Imagine a soccer game between Arsenal and West Ham. You say Arsenal has an 80% chance of winning. That could mean:

1. Long-run frequency: if the teams played many times, Arsenal would win ~8 out of 10.
2. Betting odds: you'd only take a wager if payoffs reflected an 80/20 split.
3. Subjective belief: your confidence in Arsenal is four times stronger than in West Ham.

All three interpretations follow the same rules but reflect different philosophies.

For our purposes, the math works the same either way. In this course we'll use the **frequentist** view, so you can think of probability as the long-run proportion of times an event would occur if you could repeat the process many times (that's #1 above).

4.4 Basic probability theory

At the core of probability theory is the idea of a distribution: a set of possible outcomes and their probabilities.

Suppose we track which ice cream flavor a customer chooses: chocolate, vanilla, or strawberry. Each choice is an *elementary event*, and the set of all possible events is the *sample space*, S:

$S = \text{chocolate}, \text{vanilla}, \text{strawberry}$

If we record how often each flavor is chosen, we might estimate the following probabilities:

Flavor Probability:

- Chocolate $P(\text{chocolate}) = 0.5$
- Vanilla $P(\text{vanilla}) = 0.3$
- Strawberry $P(\text{strawberry}) = 0.2$

These probabilities form a distribution. Each value lies between 0 and 1, and together they sum to 1.

Non-elementary events

We can also define events that group multiple outcomes.

For example, let $A =$ "not chocolate." Then:

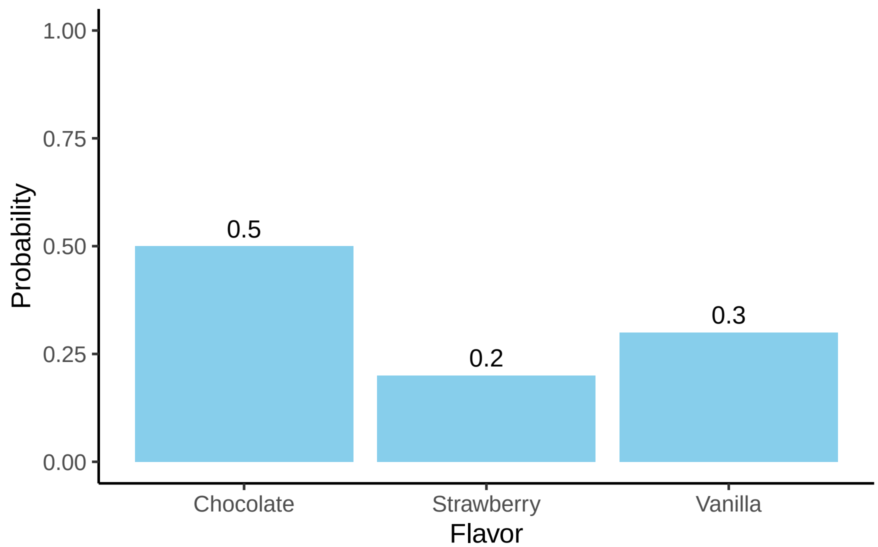

Figure 4.1: A simple probability distribution: the probabilities for three flavors must add to 1.

$$P(A) = P(vanilla) + P(strawberry) = 0.5$$

4.5 Rules of Probability

Some basic rules follow from this framework. Let's consider a sample space X that consists of elementary events x, and two non-elementary events, which we'll call A and B.

Here are our rules:

English	Notation	Formula
not A	$\neg A$	$P(\neg A) = 1 - P(A)$
A and B	$A \cap B$	$P(A \cap B) = P(A \mid B)\, P(B) = P(B \mid A)\, P(A)$
A or B	$A \cup B$	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
A given B	$A \mid B$	$P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$

These rules look abstract at first, but they are just formal ways of describing common-sense situations. Let's work through them step by step with our ice cream example. We'll assign probabilities to each flavor and then show how the rules play out with real numbers. That way, the notation is always tied to something you can picture.

Let's add a fourth flavor and assign some probabilities:

Flavor Probability:

- Chocolate $P(\text{chocolate}) = 0.4$
- Vanilla $P(\text{vanilla}) = 0.3$
- Strawberry $P(\text{strawberry}) = 0.2$
- Rocky Road $P(\text{rocky road}) = 0.1$

Notice these add to 1, as they must.

We'll have two non-elementary events, A and B, defined as:

$A = (\text{chocolate, vanilla, strawberry})$

$B = (\text{chocolate, rocky road})$

4.5.1 Not A

$A = (\text{chocolate, vanilla, strawberry})$

$P(A) = 0.4 + 0.3 + 0.2 = 0.9$

So, $P(\neg A) = 1 - P(A) = 1 - 0.9 = 0.1$

Explanation: This adds up logically. The chance of *not* choosing chocolate, vanilla, or strawberry is equal to the chance of Rocky Road.

4.5.2 A and B (the intersection)

Think of *and* as meaning "both at once." For probabilities, that translates into the **overlap** of the two events.

- The only overlap between A and B is chocolate: $A \cap B = (\text{chocolate})$
- The intersection $A \cap B$ contains only chocolate, and its probability is 0.4.
- So $P(A \cap B) = 0.4$

To reiterate: *And* means overlap, not "add the probabilities together."

4.5.3 A or B (the union)

Or in probability is inclusive: it means “in A, or in B, or in both.”

- If we combine everything in either set, we get all four flavors.
- $A \cup B =$ (chocolate, vanilla, strawberry, rocky road)
- That’s the whole sample space, so $P(A \cup B) = 1$.
- So $P(A \cup B) = 1$.

To reiterate: *or* covers everything in either group. We subtract the overlap ($A B$) so we don’t double-count it.

4.5.4 Conditional probability

Suppose we want $P(B \mid A)$: the probability of choosing B (chocolate or rocky road) given that we already know the choice was from A (chocolate, vanilla, strawberry).

By the rule,

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{0.4}{0.9} \approx 0.44$$

Explanation: Given the choice is from the A group, the only overlap with B is chocolate. So there’s about a 44% chance the flavor is chocolate.

4.6 Probability Distributions

A **probability distribution** is a rule that assigns probabilities to all possible outcomes of a random variable. Many distributions exist, but this book mostly uses five: Binomial, Normal, t, ^{2} (“chi-square”), and F. We’ll focus on Binomial and Normal here; the others appear when we do inference.

The binomial and the normal represent two kinds of distributions: **discrete** and **continuous**.

- **Discrete distributions** assign probabilities to distinct outcomes.
- **Continuous distributions** describe ranges of values, where probabilities are measured as *areas* under a curve.

4.6.1 The binomial distribution (discrete)

The binomial distribution models the number of "successes" in a fixed number of independent trials.

- A trial is a single attempt with two possible outcomes (success or failure).
- The success probability is the chance of success on any one trial, usually written as θ.
- The size parameter N is the number of trials.
- The random variable X is the number of successes observed in those N trials.

We write this as:

$X \sim \text{Binomial}(N, \theta)$

For example, if you flip a fair coin 20 times ($N = 20$, $\theta = 0.5$), the binomial distribution tells you the probability of getting 0 heads, 1 head, 2 heads … all the way up to 20 heads. The bars of the distribution in Figure **??** show the probability of each outcome, and the total always sums to 1.

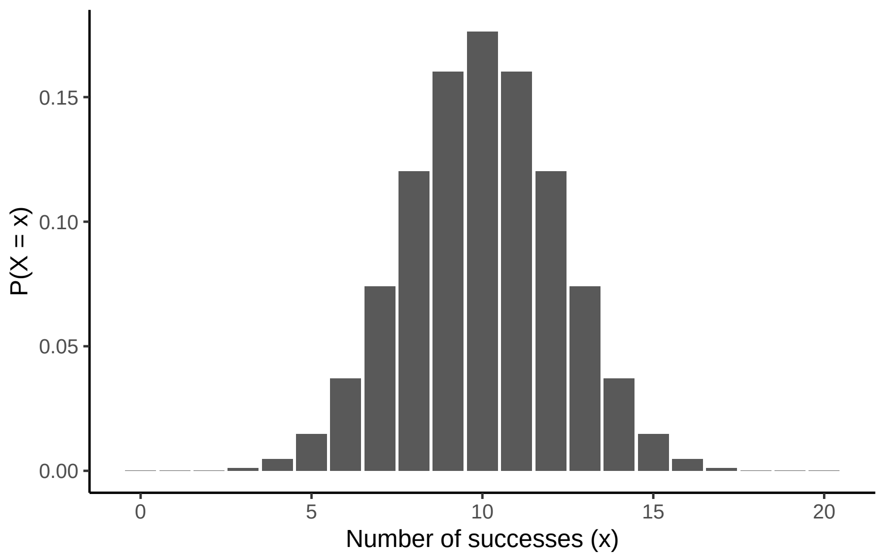

Figure 4.2: Binomial probabilities for X ~ Binomial(N = 20,   = 0.5). Bars show P(X = x).

Two common queries

1. Exact probability. For example, the chance of exactly 4 heads: $P(X = 4)$

```r
dbinom(x = 4, size = 20, prob = 0.5)
#> [1] 0.004620552
```

The probability is about 0.5%.

2. Cumulative probability. For example, $P(X \leq 4)$:

```
pbinom(q = 4, size = 20, prob = 0.5)
#> [1] 0.005908966
```

The probability is about 0.6%

The probability is slightly higher here, because it also includes the probability we could get 3, 2, or 1 heads as well.

> **Note**
>
> R's distribution helpers (for every distribution):
>
> - d* = exact probability (PDF)
> - p* = cumulative probability $P(X \leq q)$
> - q* = quantile (inverse of p*)
> - r* = random draws
>
> For binomial: dbinom, pbinom, qbinom, rbinom. For discrete distributions, some percentiles don't exist exactly; qbinom returns the smallest x with P(X x) p.

4.6.2 The normal distribution (continuous)

The normal distribution describes variables that can take on a wide range of values, clustered around a central point. The normal distribution is our most important distribution. It's also sometimes called a Gaussian distribution or a bell curve.

- The mean μ is the center of the distribution.
- The standard deviation σ measures spread: how tightly or widely values are clustered around the mean.
- The random variable X represents the outcome, which can be any real number.

We write this as:

$X \sim \text{Normal}(\mu, \sigma)$

It looks like Figure **??**.

> **Note**
>
> For continuous distributions: - Point probability = 0 - Probability = area under curve

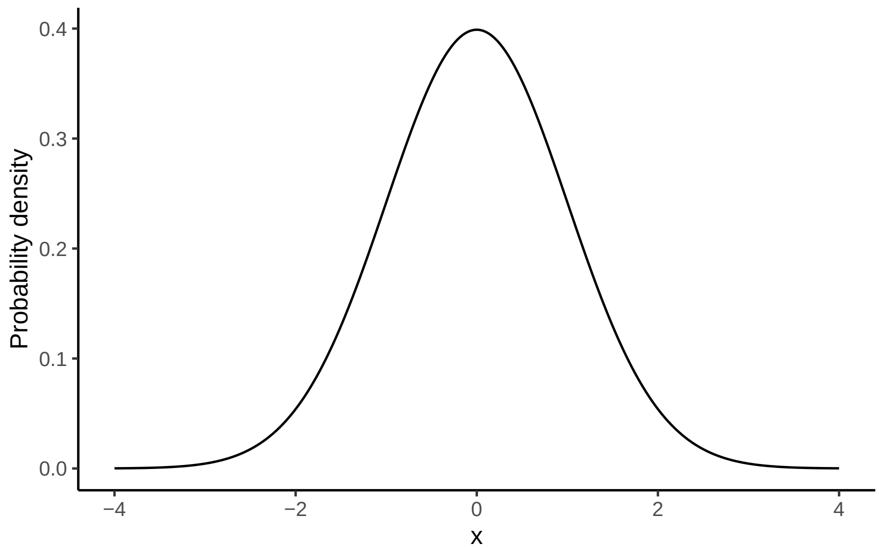

Figure 4.3: Standard normal Probability Distribution Function (PDF), where = 0, = 1. For continuous variables, probabilities are *areas* under the curve.

Changing μ shifts the curve left/right; changing σ widens/narrows it (area always = 1).

For continuous distributions $P(X = x) = 0$; instead, we compute $P(a \leq X \leq b)$ as the area between a and b.

See Figure **??** for an illustration.

This leads to a key property of the normal distribution (explored further in the Z-scores section):

68–95–99.7 rule:

- ~68% of values fall within 1 of the mean
- ~95% fall within 2
- ~99.7% fall within 3

Two common queries

1. Probability of being below a value, e.g. $P(X \leq 1.64)$ for $X \sim \text{Normal}(0, 1)$

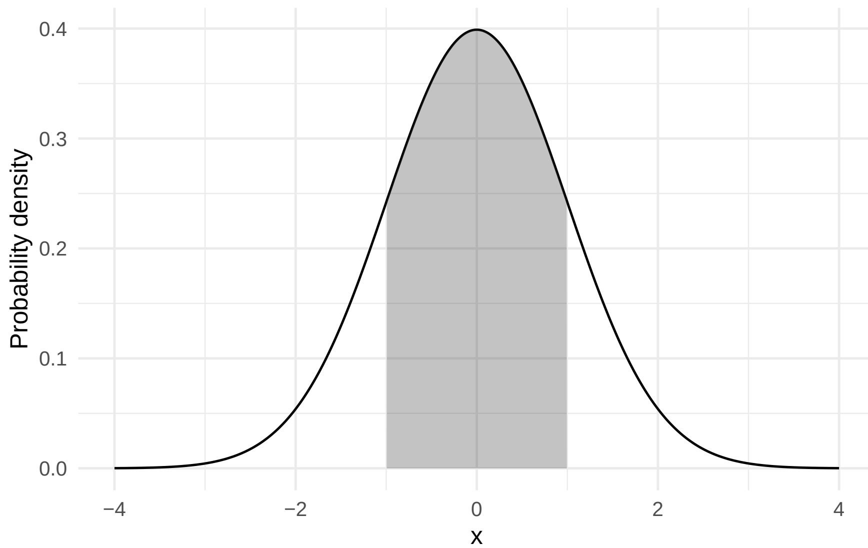

Figure 4.4: Shaded area shows P(−1 X 1) 0.68 for X ~ Normal(0,1).

```
pnorm(q = 1.64, mean = 0, sd = 1)
#> [1] 0.9494974
```

In a normal distribution with a mean of 0 and standard deviation of 1, the probability the value is less than or equal to 1.64 is about 95%.

2. Value for a percentile, e.g. the 97.5th percentile:

```
qnorm(p = 0.975, mean = 0, sd = 1)
#> [1] 1.959964
```

Takeaway: For continuous variables, use areas (via pnorm/qnorm), not bar heights.

4.6.3 What's probability density?

For continuous distributions like the normal, probabilities work differently than in the discrete case. You cannot ask for the probability of one exact value (e.g., $P(X = 23)$), because the answer is essentially zero. Instead, we calculate the probability of being in a **range**, such as $22.5 \leq X \leq 23.5$.

This is why the y-axis of a normal curve is labelled **probability density**. The height of the curve at a point, $p(x)$, is not itself a probability. Instead, probabilities come from the **area**

under the curve between two values. For example, about 68% of values from a standard normal fall between -1 and 1, because the area between -1 and 1 is 0.68.

In R, functions like `dnorm()` return the density (the curve's height), while `pnorm()` and `qnorm()` work with cumulative probabilities and quantiles, which are usually more useful for applied problems.

4.7 Summary of Probability

In this section we introduced the building blocks of probability, with an emphasis on the parts most relevant for statistics.

- What probability means: We use the frequentist view—probability as the long-run proportion of times an event occurs.
- Distributions: A probability distribution assigns probabilities to all possible outcomes of a random variable. For discrete variables (like coin flips), probabilities are attached to each outcome; for continuous variables (like exam scores), probabilities come from areas under a curve.
- Rules of probability: Complements, intersections (and), unions (or), and conditional probabilities all follow from the basic distribution framework.
- Key distributions:
  - Binomial—counts successes across N independent trials.
  - Normal—describes continuous outcomes centered at a mean, with spread determined by the standard deviation.
  - Others (t, χ^2, F) appear later when we turn to inference.

Takeaway: probability provides the language and tools that make statistical inference possible. Once we understand how outcomes behave under these distributions, we can ask questions about samples, uncertainty, and whether results are surprising enough to change what we believe about the world.

4.8 Samples, populations and sampling

Descriptive statistics summarize what we **do** know. Inferential statistics aim to "learn what we do not know from what we do." The central questions are: what do we want to learn about, and how do we learn it? Introductory statistics typically divides inference into two big ideas: estimation and hypothesis testing. This chapter introduces estimation, but first we need to cover sampling. Estimation only makes sense once you understand how samples relate to populations.

Sampling theory specifies the assumptions on which statistical inferences rest. To talk about making inferences, we must be clear about what we are drawing inferences **from** (the sample) and what we are drawing inferences **about** (the population).

In almost every study, what we actually have is a **sample**—a finite set of data points. We cannot survey every voter in a country or measure every tree in a forest. Earlier, our goal was just to describe the sample. Now we turn to how samples can be used to generalize.

4.8.1 Defining a population

A sample is tangible: it is the dataset you can open on your computer. A **population**, by contrast, is the (usually much larger) set of all possible individuals or observations you want to draw conclusions about. In an ideal world, researchers would start with a clear definition of the population of interest, since that shapes how a study is designed. In practice, the population is often only loosely defined. Still, the basic idea is simple: the sample should represent some broader population we care about.

4.8.2 Simple random samples

The sample–population relationship depends on the **procedure** used to select the sample. This is the **sampling method**, and it matters.

Suppose we have a bag with 10 chips, each uniquely labeled, some black and some white. This bag is the population. If we shake the bag and draw 4 chips without replacement, each chip has the same chance of being selected. This is a **simple random sample**.

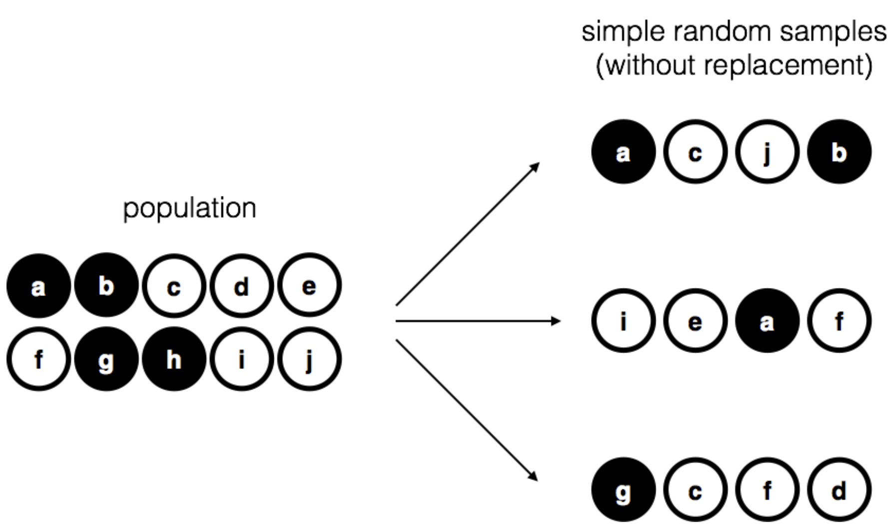

Figure 4.5: Simple random sampling without replacement from a finite population

Now imagine a different procedure: someone peeks inside and deliberately picks only black chips. That is a **biased sample**. The difference is critical. With a random sample, we can use statistical tools to generalize from the sample to the population; with a biased sample, we cannot.

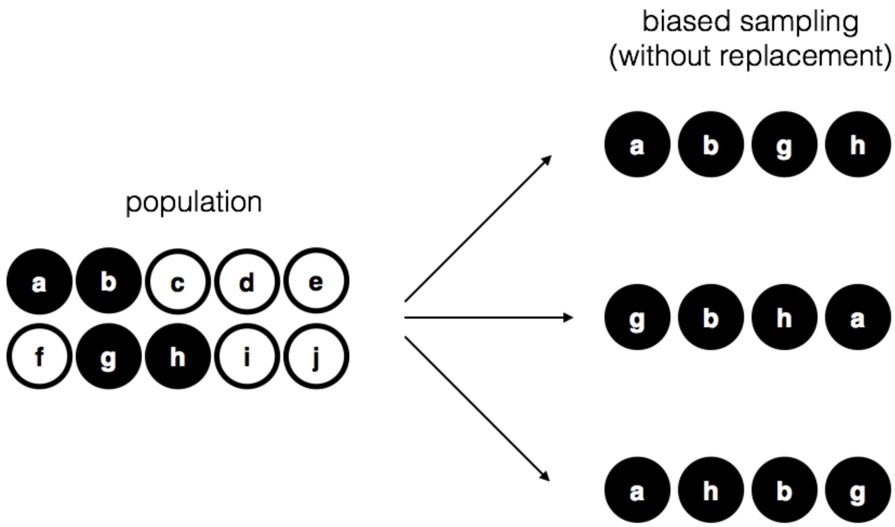

Figure 4.6: Biased sampling without replacement from a finite populations.

Another variation is to draw chips **with replacement**—each chip is returned to the bag before the next draw. In practice, most studies are "without replacement," but much of statistical theory assumes replacement. When the population is large, the difference is negligible.

4.8.3 Random and non-random samples in practice

Random samples are the gold standard, and in many areas of environmental science they are common: inventory plots, transects, and monitoring networks are usually designed with probability-based methods. Still, many studies rely on modified approaches:

- **Stratified sampling** divides the population into subgroups (strata) and samples within each. This ensures representation of small but important groups, such as rare habitat types.
- **Opportunistic or logistically constrained sampling** selects cases that are accessible or already monitored. These are not random, but they are common in practice when budgets, terrain, or historical networks limit where data can be collected. .

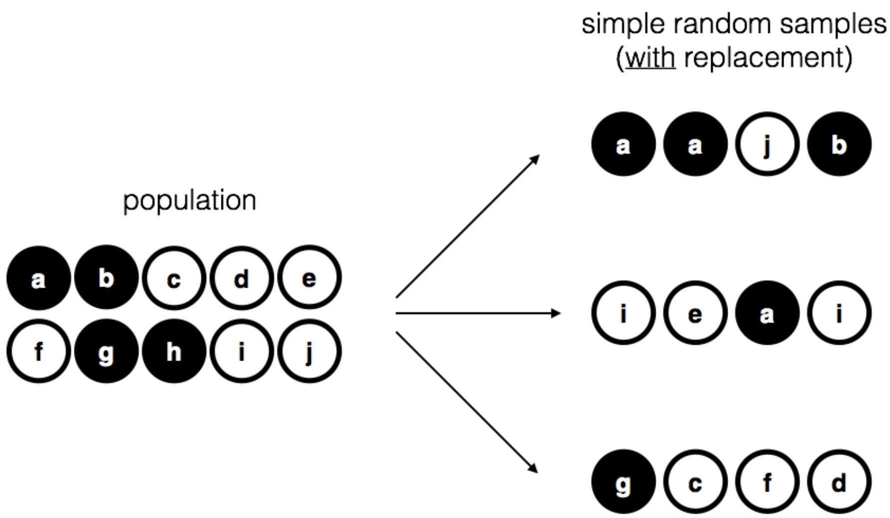

Figure 4.7: Simple random sampling with replacement from a finite population.

4.8.4 How much does it matter if you don't have a random sample?

The short answer: it depends. Stratified sampling, while biased by design, is often useful and correctable with statistical adjustments. Opportunistic samples can be acceptable if the bias is unlikely to affect the phenomenon being studied. For example, if you want to study tree growth rates, sampling only trees on one side of a valley may or may not be problematic, depending on whether the valley side influences growth.

The key is to consider whether the way you sampled could plausibly distort the conclusions. When designing your own studies, aim for sampling methods that are as appropriate as possible for the population of interest. And when critiquing others, be specific about how a sampling choice might bias results, rather than simply noting that it is not random.

4.8.5 Population parameters and sample statistics

So far we have talked about populations the way a scientist might: a group of people, a stand of trees, a set of rivers. Statisticians formalize this idea by treating a population as a **probability distribution**—the abstract source from which data are drawn.

For example, suppose daily coffee consumption in adults follow an approximately normal distribution with mean $\mu - 2$ cups/day and standard deviation $\sigma = 1$ cup/day. These are the **population parameters**.

Now suppose we take a random sample of 100 adults. The histogram in panel (b) of Figure **??** shows that the sample resembles the population distribution, but not perfectly. The sample

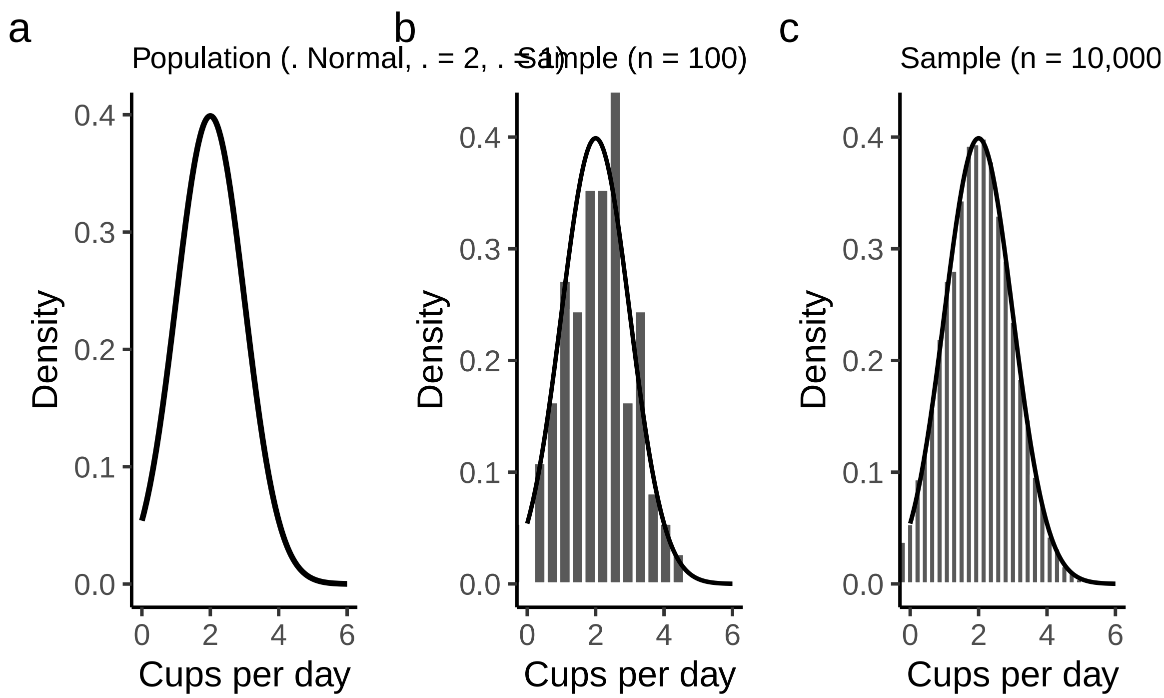

Figure 4.8: The population distribution of daily coffee consumption (a) and two random samples: one of size 100 (b) and one of size 10,000 (c).

mean might be 1.9 cups and the sample standard deviation 1.1—close to, but not exactly, the population values.

These are **sample statistics**: numbers we calculate from the data in hand. They approximate the population parameters, which describe the entire distribution. The central problem of inference is how to use sample statistics to estimate population parameters, and how much confidence we can place in those estimates.

4.9 The law of large numbers

In the last section we compared a small sample of coffee drinkers ($n = 100$) to a much larger sample ($n = 10{,}000$). The difference was clear: the larger sample produced a histogram that looked much closer to the true population distribution. The sample mean and standard deviation were also much closer to the population values ($\mu = 2$, $\sigma = 1$ cups/day).

This illustrates the **law of large numbers**. Informally: larger samples give better information. More precisely, as the sample size grows, the sample mean $\bar{X}$ converges toward the population mean μ. Symbolically:

$$N \to \infty \quad \Rightarrow \quad \bar{X} \to \mu$$

Although easiest to see with the mean, the law applies to many sample statistics: proportions, variances, correlations, and more. The guarantee is that, with enough data, the statistics you calculate from a sample will hover ever closer to their population counterparts.

The idea is obvious enough that Jacob Bernoulli, who first formalized it in 1713, remarked that "even the most stupid of men" already know it by instinct. His phrasing hasn't aged well, but the insight remains correct: averaging over more observations reduces the "danger of wandering from one's goal."

In practice, this means any single sample statistic will be off the mark, but if we keep collecting data, those statistics will tend to get closer and closer to the true population parameters. The law of large numbers is one of the fundamental guarantees that makes statistical inference possible.

4.10 Sampling distributions and the central limit theorem

The law of large numbers tells us that with enough data, our sample mean will get close to the population mean. But in practice, we never have infinite data. What we need is a way to understand how the sample mean behaves with *finite* samples. This is where the **sampling distribution of the sample means** comes in.

4.10.1 Sampling distribution of the sample means

The phrase is clunky, but the idea is straightforward.

- A **sample mean** is just the average of one sample.
- A **sampling distribution** is what you get when you take many samples, compute the mean for each, and then look at the distribution of those means.

So the **sampling distribution of the sample means** is simply the distribution formed by repeating your study many times and collecting all those sample averages.

Why is this useful? Because it tells us how much the sample mean varies from sample to sample, and how close we can expect it to be to the population mean.

4.10.2 Seeing the pieces

We can build this step by step:

1. Start with a distribution to sample from.
2. Draw repeated random samples of the same size n.
3. Compute the mean of each sample.
4. Plot those means in a histogram.

For example, suppose we draw numbers from a uniform distribution on 1 to 10: Figure **??**.

Individual samples of size 20 look very different from each other, but their means cluster near 5.5 (the population mean).

If we repeat this thousands of times and plot all the sample means, we get a new distribution: the sampling distribution of the sample means. It is centered on 5.5, but spread out because of sampling variation.

We had a uniform distribution originally, but Figure **??** does not look like a uniform distribution. This is key takeaway: the histogram of sample means does **not** look like the original uniform distribution. Instead, it has its own shape and variability. That fact leads directly to the **central limit theorem**, which we'll introduce next.

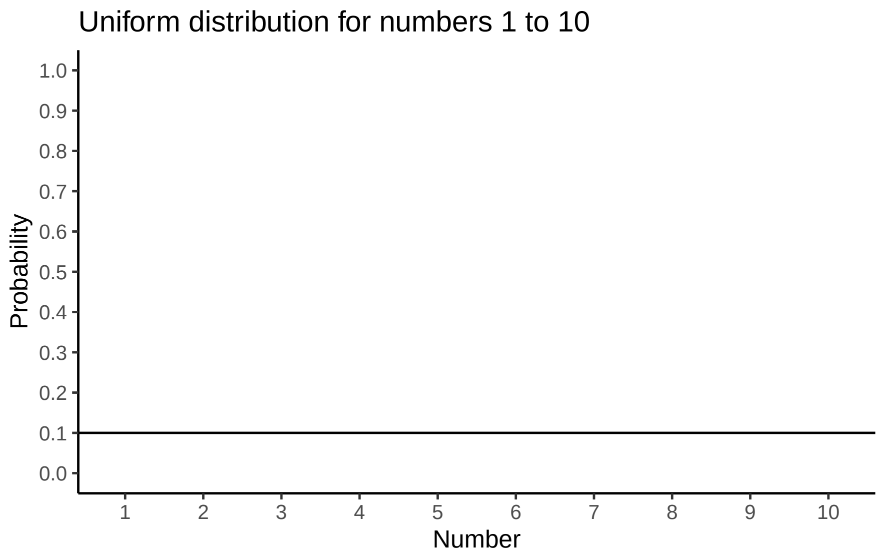

Figure 4.9: A uniform distribution illustrating the probabilites of sampling the numbers 1 to 10. In a uniform distribution, all numbers have an equal probability of being sampled, so the line is flat indicating all numbers have the same probability

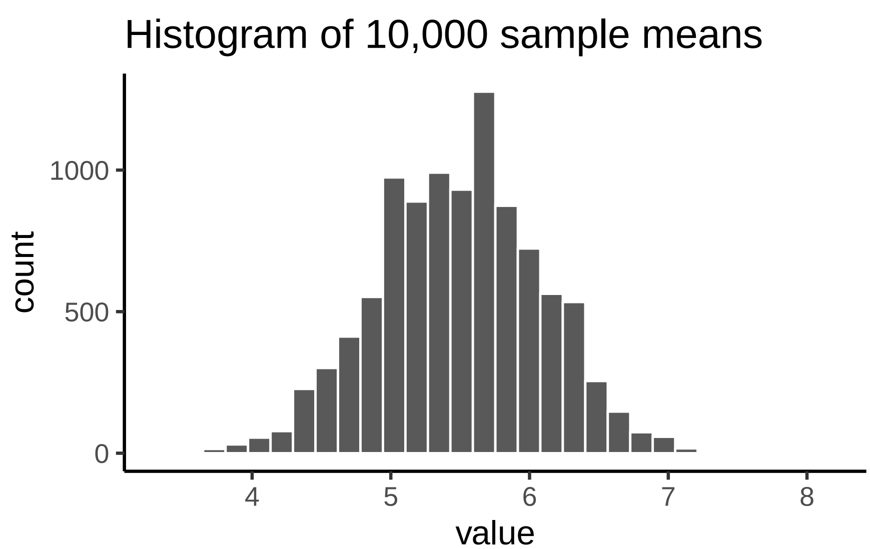

Figure 4.10: A histogram showing the sample means for 10,000 samples, each size 20, from the uniform distribution of numbers from 1 to 10. The expected mean is 5.5, and the histogram is centered on 5.5. The mean of each sample is not always 5.5 because of sampling error or chance

4.10.3 Beyond the mean

Although we've focused on means, the same idea applies to any statistic—medians, standard deviations, maximum values, and so on. Each has its own sampling distribution, which describes how that statistic behaves across repeated samples. To illustrate, suppose we draw many samples of size $n = 50$ from a normal distribution with mean 100 and standard deviation 20. For each sample we compute the mean, standard deviation, maximum, and median, then plot the histograms of these values (Figure **??**).

The sample mean is just the most important case, because it underpins much of classical inference.

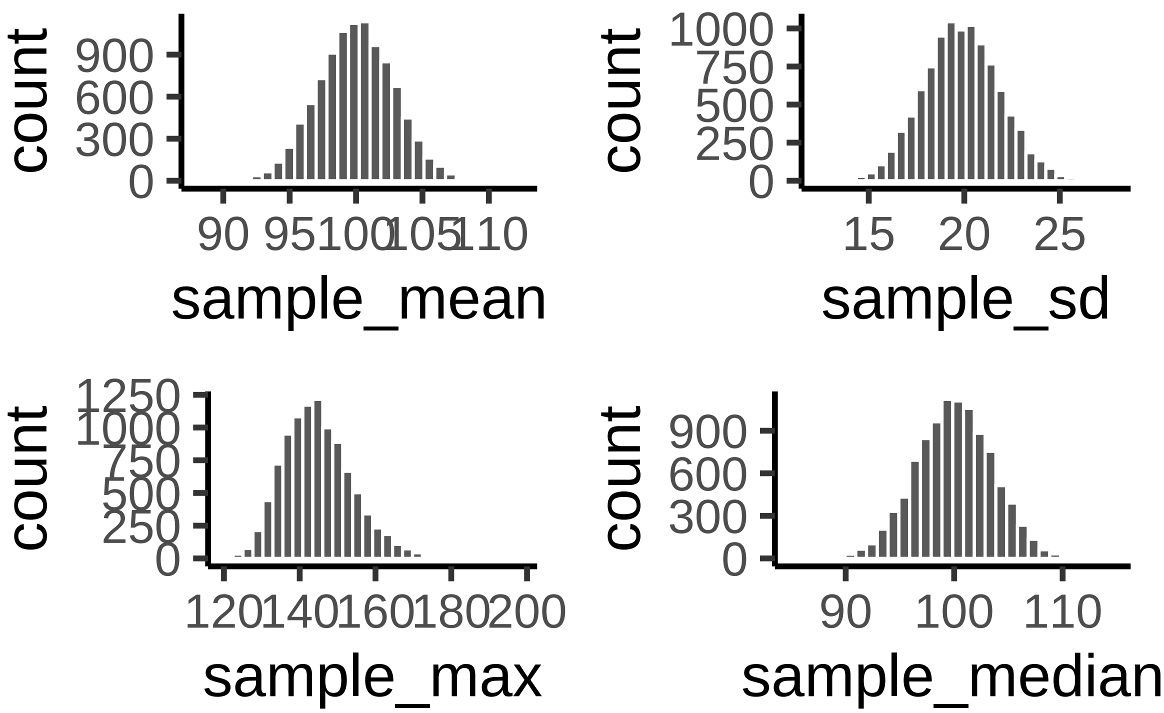

Figure 4.11: Each panel shows a histogram of a different sampling statistic

4.11 The central limit theorem

So far, you've seen that sample means vary from one sample to the next, and that their variability shrinks as sample size grows. The **sampling distribution of the mean** is just the distribution you get if you compute the mean from many different samples of the same size.

Intuitively: - Small samples → sample means bounce around more → wide sampling distribution. - Large samples → sample means are more stable → narrow sampling distribution.

?@fig-4samplingmean animates this idea. Each panel shows samples of size 10, 50, 100, or 1000 drawn from a normal distribution (red line). Grey bars show the sample itself, the red line marks that sample's mean, and the blue histogram shows the distribution of sample means across many repetitions. Notice how the blue distribution shrinks around the population mean as n increases.

So, the sampling distribution of the mean is itself a distribution, with its own variability. When the sample size is small, the distribution of sample means is wide; when the sample size is large, it is narrow. The standard deviation of this sampling distribution has a special name: **the standard error (SE)**. In this context, it is the standard error of the mean. As you can see in the animation, as the sample size N increases, the SE decreases.

But there's more. The **central limit theorem** tells us three powerful facts about the sampling distribution of the mean:

1. Its mean equals the population mean, μ.
2. Its standard deviation equals the population σ divided by $\sqrt{N}$:

$$\text{SEM} = \frac{\sigma}{\sqrt{N}}$$

3. Its shape approaches normal as N grows — no matter what the shape of the population is.

Together, these three facts form our practical version of the **central limit theorem (CLT)**. The first fact, that sample means converge to the population mean, comes from the law of large numbers. The second formalizes how the SE shrinks with larger N. And the third shows why the normal curve appears so often: the distribution of sample means tends toward normality, even when the underlying data are skewed or flat.

The full CLT is mathematically more general and has a complicated proof, but for our purposes these three results capture what we need. To see the third point in action, let's look at histograms of **sample means** drawn from different underlying distributions. Remember: these are distributions of *sample means*, not of individual observations.

In Figure **??**, we are smapling from a *normal* distribution (red line). The sampling distribution of the mean (blue bars) is also normal.

Let's try sampling from a uniform population: the parent distribution is flat (red), but the sampling distribution of the mean is bell-shaped (Figure **??**).

Sampling from an exponential population: the parent distribution is skewed, but the sampling distribution of the mean still looks normal (Figure **??**).

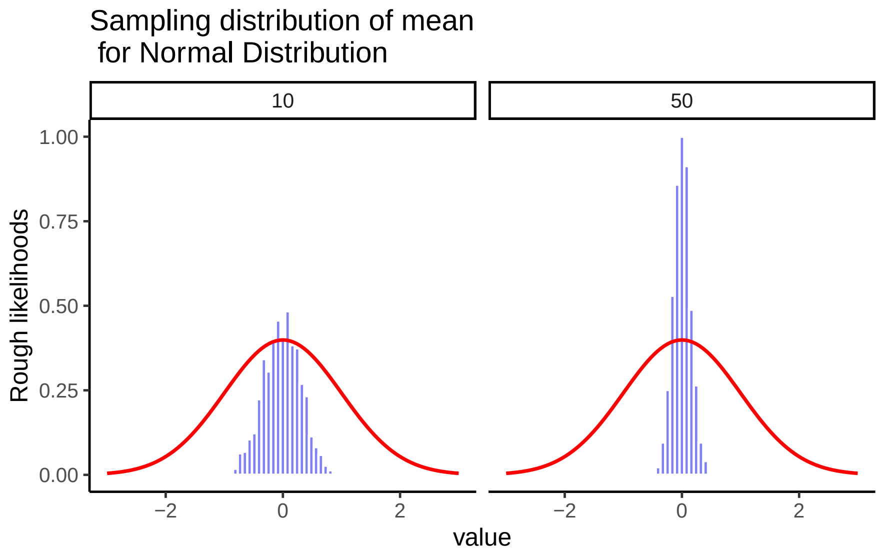

Figure 4.12: Sampling distributions of the mean from a normal population. As sample size increases, the distribution tightens around the population mean.

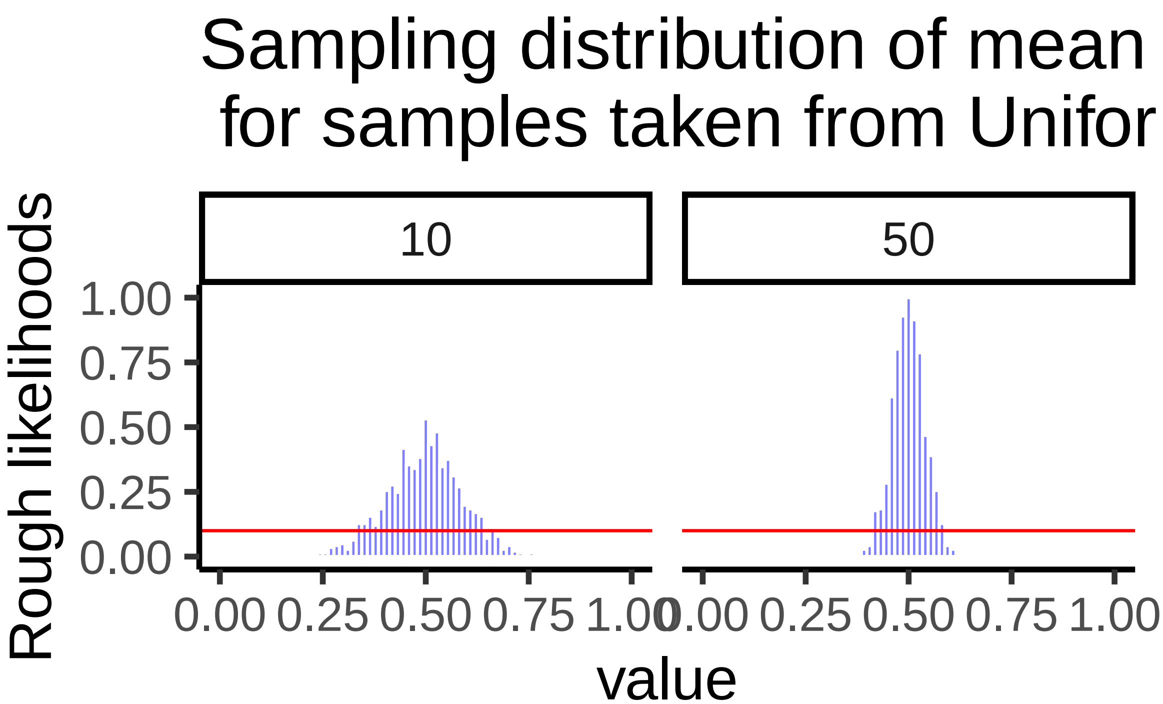

Figure 4.13: Even when samples come from a flat uniform distribution, the distribution of sample means is approximately normal.

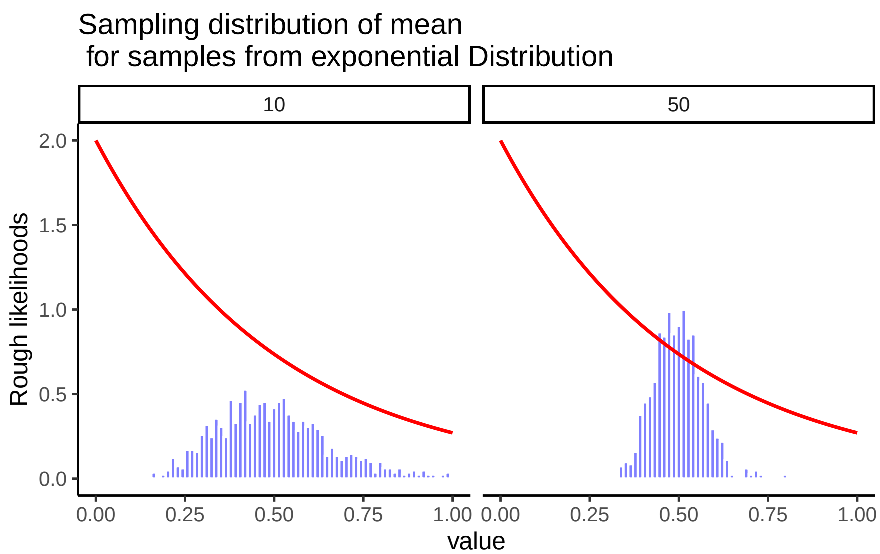

Figure 4.14: Even when samples come from a skewed exponential distribution, the distribution of sample means is approximately normal.

Together, these results are the essence of the central limit theorem (CLT). Formally, if the population has mean μ and standard deviation σ, then the sampling distribution of the mean also has mean μ and standard error

$$SE = \frac{\sigma}{\sqrt{N}}.$$

This formula shows how sample size controls reliability. As N increases, the denominator grows and the SE falls. The drop is not linear: doubling N cuts the SE by only about 30%, which is why the benefits of ever-larger samples taper off. Still, larger samples consistently produce more stable and trustworthy estimates of μ.

The CLT is one of the cornerstones of statistics. It explains why sample means are useful, why bigger studies are more reliable, and why the normal curve appears so often in practice. Whenever you average across many influences, exam scores, crop yields, daily temperatures, the distribution of those averages tends to be normal.

4.12 z-scores

A **z-score** is a standardized way to express how far a value is from its mean, measured in units of standard deviation. Z-scores are useful because (i) areas under the normal curve